

SMART VISION-BASED ATTENDANCE SYSTEM USING IOT AND ML WITH REAL TIME NOTIFICATIONS

A Project Report

Submitted to the FACULTY of ENGINEERING of

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, KAKINADA

in partial fulfillment of the requirements,

for the award of the Degree of

Bachelor of Technology

in

Internet of Things

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SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE
(An autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

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ANDHRA PRADESH

2025-2026

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CERTIFICATE

This is to certify that the project report entitled “**SMART VISION-BASED ATTENDANCE SYSTEM USING IOT AND ML WITH REAL TIME NOTIFICATIONS**” is a bonafide work carried out by **Sowmya Sri.G (22481A60A2), M.Mohammad Abbas (22481A6068), T.Srikanth (22481A60A4), M.Bhaviya Sri (23485A6014)** under my guidance and supervision in partial fulfilment of the requirements, for the award of the degree of Bachelor of Technology in **Internet of Things** of Seshadri Rao Gudlavalleru Engineering College affiliated to **Jawaharlal Nehru Technological University, Kakinada.**

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NOMENCLATURE

IoT	-	Internet of Things
ML	-	Machine Learning
LCD	-	Liquid Crystal Display
GSM	-	Global System for Mobile Communications
RFID	-	Radio Frequency Identification
LBPH	-	Local Binary Pattern Histogram
LBP	-	Local Binary Pattern
SMS	-	Short Message Service
GPIO	-	General Purpose Input Output
CSI	-	Camera Serial Interface
DSI	-	Display Serial Interface
HDMI	-	High Definition Multimedia Interface
SOC	-	System on Chip
CMOS	-	Complementary Metal-Oxide Semiconductor
RAM	-	Random Access Memory
ROM	-	Read Only Memory
SD	-	Secure Digital Card
OpenCV	-	Open Source Computer Vision
IP	-	Internet Protocol
I ² C	-	Inter-Integrated Circuit
SPI	-	Serial Peripheral Interface
GUI	-	Graphical User Interface

ABSTRACT

In Educational Institutions and Professional Settings, maintaining accurate attendance records is vital for monitoring performance, ensuring security, and improving overall management. Traditional methods like manual attendance or RFID-based systems are time consuming, prone to proxy attendance, and require physical interaction, which is inefficient and outdated in today's tech-driven environment. These methods often fail to provide real time updates and are not scalable for large institutions.

The Smart Vision-Based Attendance System addresses the limitations of traditional attendance methods by using IoT technologies and Machine Learning (ML) to automate and streamline the attendance process. A camera with ML algorithms captures and recognizes the faces of students as they enter the classroom, automatically marking their attendance and storing it in a cloud database in real time. If a student is found absent, the system sends a notification to the parent or guardian, while administrators can monitor attendance records through a connected web or mobile dashboard. This provides a contactless, efficient, and scalable solution for modern attendance management, enhancing accuracy, student safety, and overall institutional productivity.

Keywords: Vision-Based Attendance System, Face Recognition, LBPH, IoT, Cloud Based Monitoring, Real Time Notification System.

CHAPTER 1

INTRODUCTION

1.1 Background:

Face recognition plays a vital role in daily life, allowing humans to identify family, friends or familiar individuals. This process, while seemingly effortless, involves multiple complex steps. Human intelligence receives visual information through the retina in the form of light, which is a type of electromagnetic wave projected from a source onto an object. As explained by Robinson-Riegler and Robinson-Riegler (2008), the human visual system processes this information by analyzing shape, size, contour, texture and then compares it with stored representations in memory to recognize faces accurately.

However, replicating this human ability in an automated system is a significant challenge. Humans have limited memory and can easily forget or confuse faces, especially in environments with a large number of individuals, such as universities with students of diverse races and genders. To overcome these limitations, computers are employed in face recognition systems, leveraging high processing power, rapid computation, and near-limitless memory to identify faces accurately.

The human face is a unique representation of individual identity. Therefore, face recognition is considered a biometric method, where an individual is identified by comparing a real-time captured image with pre stored images in a database (Margaret Rouse, 2012). This technology forms the foundation of modern Smart Vision-Based Attendance Systems, which combine IoT and Machine Learning to automate attendance tracking while providing real-time notifications to ensure efficiency, accuracy and security.

1.2 Aim of this Project:

The aim of this project is to develop a real-time, smart attendance management system that integrates a camera with IoT-enabled devices, leveraging Machine Learning-based face recognition algorithms for accurate and efficient student identification. The system is designed to function in classrooms and professional settings, automatically capturing and recognizing faces as individuals enter and marking attendance without manual intervention.

Beyond simple attendance tracking, the system provides real-time notifications to parents or guardians if a student is absent and allows administrators to monitor records through a web or mobile dashboard, ensuring transparency and timely updates. The project also aims to optimize performance by processing recognition in real time and storing attendance data securely in a cloud database.

By combining AI-driven face recognition, automated notifications and cloud-based monitoring, this project seeks to create a scalable, contactless and efficient solution for attendance management, enhancing accuracy, student safety and institutional productivity.

1.3 Methodology:

The technical approach to developing a Smart Vision-Based Attendance System integrates both hardware and software components to achieve real-time face recognition and attendance tracking. The system uses a camera connected to an IoT-enabled device for capturing live video, enabling contactless and automated monitoring of students or employees.

The software implementation is built using Machine Learning-based face recognition algorithms, such as Local Binary Pattern Histogram (LBPH), trained to accurately identify individuals by comparing captured images with pre-stored records in the database. The image processing pipeline continuously streams live video from the camera to the processing unit. Frames are extracted at regular intervals and analyzed to detect and recognize 8 faces in real time.

A time-based alert system triggers notifications via email, SMS or other methods if human presence is detected during predefined critical periods. This feature enhances security and surveillance capabilities. All detected objects are logged for future analysis and annotated images are stored or displayed for real-time monitoring. The project demonstrates a comprehensive approach to surveillance and object detection, leveraging the strengths of both hardware and software components. Its applications extend to various industries, including security, law enforcement, and smart home automation.

The system ensures high accuracy and efficiency by combining real-time processing with reliable database management. It also supports scalable deployment, allowing integration with cloud platforms and centralized monitoring systems. Overall, the system provides a secure and smart attendance solution for modern environments.

1.4 Significance of this Work:

The Smart Vision-Based Attendance System plays a crucial role in advancing automated attendance management by integrating IoT-enabled cameras with Machine Learning-based face recognition for real-time identification. By leveraging AI algorithms, the system ensures accurate recognition of individuals, making attendance tracking contactless, efficient and reliable in classrooms, offices or professional environments.

One of the most significant aspects of this project is its real-time notification mechanism, which alerts parents, guardians or administrators if a student or employee is absent. This feature enhances accountability and ensures timely communication, reducing delays and manual follow-ups. Automated alerts also minimize human intervention while maintaining transparency and oversight.

Additionally, the system's cloud-based storage and dashboard monitoring improve situational awareness, allowing administrators to access, analyze and manage attendance records efficiently. Its scalable and cost 9 effective design makes it a practical alternative to traditional attendance systems, which often involve manual entry or RFID-based processes that are time-consuming and error-prone. By integrating AI-driven face recognition, IoT connectivity and cloud monitoring, the project significantly enhances accuracy, reduces human effort and modernizes attendance management for educational and professional institutions.

1.5 Outline of this Report:

This Report is organized as follows:

- Chapter 1 introduces the background, aim, methodology and significance of the project.
- Chapter 2 presents the literature review of the existing attendance systems.
- Chapter 3 explains the system design and architecture.
- Chapter 4 deals with Hardware Implementation.
- Chapter 5 deals with Software Implementation.
- Chapter 6 deals with Results.
- Chapter 7 deals with Conclusion and Future Scope.

1.6 Conclusion:

This chapter highlights a Smart Vision-Based Attendance System that offers a powerful solution for real-time, contactless attendance tracking. Its ability to recognize individuals automatically, provide real-time notifications, and maintain cloud-based records enhances institutional efficiency, reduces manual effort, and ensures the safety and accountability of students and employees. The system is scalable, cost-effective, and suitable for educational institutions and professional settings, providing a modern approach to attendance management.

CHAPTER 2

LITERATURE REVIEW

[1] A. Kumar et al. (2022) develop an IoT-enabled real-time face recognition attendance system using lightweight machine learning models. Their method enhances speed and resource efficiency, though accuracy may reduce under poor lighting conditions.

[2] M. Naveen Raj and R. Vadivel (2023) present a machine learning-based face recognition attendance system with location identification. The system ensures secure and location-aware attendance tracking, but implementation requires reliable GPS and image datasets.

[3] G. Ramani et al. (2023) present a Smart Attendance System using IoT technologies for centralized attendance management. The system automates data storage and generates reports for administrative use. The research highlights the benefits of paperless systems and real-time data access.

[4] N. Ramakrishnan et al. (2023) propose a Wi-Fi Based Smart Attendance Monitoring System that tracks student presence through network connectivity. The system detects devices connected to institutional Wi-Fi to mark attendance automatically. While the approach reduces hardware costs, it cannot uniquely identify individuals and may produce inaccurate results in crowded environments.

[5] A. R. Ansari and L. Ahmad (2023) present a smart attendance system using deep neural networks and edge computing. The approach improves real-time processing and reduces latency, but requires advanced hardware support.

[6] Preetham B R, Prajwal R & Sanjay K (2023) present a Smart Attendance System using RFID and Face Recognition technology. The study integrates dual authentication mechanisms to enhance attendance accuracy and prevent proxy attendance. RFID cards are used for quick identification, while face recognition ensures biometric verification. The system improves reliability and reduces manual errors in attendance management. However, the integration of multiple technologies increases system complexity and cost. The research emphasizes the importance of combining biometric and automated systems to enhance institutional efficiency.

[7] K. S. Reddy et al. (2023) propose an IoT-based real-time face recognition attendance system using classical machine learning techniques. The system integrates IoT devices with face recognition to automate attendance, improving accuracy and efficiency but remaining sensitive to lighting and dataset quality.

[8] M. R. Islam and M. S. Ahmed (2024) present a vision-based real-time attendance system designed with weather-resistant cameras for outdoor environments. The system ensures reliable face detection and recognition even in outdoor conditions by using robust imaging hardware and real-time processing. Although the system enhances usability in open spaces, environmental factors such as extreme weather and partial occlusions can still impact recognition accuracy.

[9] N. Thaleeparambil, A. Biju & B. Prathap (2024) propose an Integrated RFID-WiFi-Vision Attendance System designed to improve automation in smart classrooms. The system combines RFID tagging, Wi-Fi connectivity, and vision-based recognition to ensure accurate attendance recording. The research highlights real-time data synchronization and remote monitoring capabilities. While the system enhances flexibility and monitoring efficiency, it requires strong network infrastructure and involves higher setup complexity.

[10] T. Abderraouf et al. (2024) introduce an embedded intelligent monitoring system using AI techniques. The approach focuses on automated detection and system efficiency, though embedded deployment may increase hardware complexity.

[11] S. Das and A. Mukherjee (2024) propose a smart classroom attendance system combining AI and IoT with real-time alerts. The system improves monitoring and instant notifications; however, integration increases system cost and configuration effort.

[12] D. V. Darshan et al. (2024) introduce an IoT-Based Attendance Monitoring System with Facial Recognition for real-time student tracking. The study integrates cloud-based dashboards and IoT devices to enable administrators to monitor attendance remotely. The system automatically captures student images and updates attendance records in real time. Although the approach improves transparency and efficiency, facial recognition performance is affected under poor lighting conditions and occlusions.

[13] A. Raj et al. (2024) develop a Facial Recognition-Based Student Attendance System focused on contactless biometric identification. The research demonstrates how real-time face detection algorithms can eliminate manual attendance processes and enhance classroom productivity. The system ensures secure identity verification and minimizes human errors. However, high computational requirements pose challenges for smooth real-time processing.

[14] O. Kale et al. (2025) propose a real-time facial recognition-based attendance system integrated with a web platform. The system automates attendance recording and

enables centralized data access; however, performance depends on network stability and image quality.

[15]. D. K. B. et al. (2025) develop a real-time student attendance system using AI-based face recognition. The model enhances automation and accuracy, though recognition efficiency may vary with environmental conditions.

The literature review reveals significant advancements in smart attendance systems through the integration of IoT, artificial intelligence, deep learning and edge computing technologies. Existing research highlights the benefits of automation, real-time monitoring, biometric authentication and cloud-based data management in improving attendance accuracy and institutional efficiency. Studies also emphasize challenges such as high computational requirements, lighting sensitivity, network dependency and system cost. These findings provide a strong foundation for developing a scalable, secure, and real-time IoT-based face recognition attendance management system.

CHAPTER 3

SMART VISION-BASED ATTENDANCE SYSTEM USING IOT AND ML WITH REAL TIME NOTIFICATIONS

3.1 Existing Work:

The existing work in attendance management systems focuses on reducing manual effort and improving accuracy through various identification technologies. Traditionally, attendance was recorded manually using paper registers, where teachers or administrators marked student presence. Although simple to implement, this method was time-consuming, prone to human errors and vulnerable to proxy attendance. Maintaining and analyzing records also required significant effort, making the process inefficient for large institutions.

With technological advancements, RFID-based attendance systems were introduced, where students used ID cards embedded with RFID tags to mark their attendance. These systems improved speed and digital record maintenance but were still susceptible to misuse, as students could exchange cards and mark attendance for others. Fingerprint-based biometric systems were later adopted to enhance security. While they provided better authentication, they required physical contact and one-by-one verification, leading to delays and hygiene concerns in crowded environments.

Iris and voice recognition systems were also explored to increase accuracy and uniqueness in identification. Iris recognition offered highly detailed biometric data but raised privacy concerns and required expensive hardware. Voice recognition systems were easier to implement but showed lower accuracy due to background noise and variations in speech patterns. These limitations in existing systems highlighted the need for a contactless, efficient, and automated solution, leading to the development of vision-based face recognition attendance systems for real-time and reliable monitoring.

However, many early face recognition systems struggled with variations in lighting, facial expressions and pose angles. Some systems also lacked real-time notification features and centralized data storage. Scalability and integration with IoT platforms were limited in earlier implementations. These gaps created the foundation for developing a more robust Smart Vision-Based Attendance System with IoT and real-time alert capabilities. Additionally, improvements in machine learning algorithms and cloud connectivity have enabled faster processing and better accuracy.

Table 3.1: Advantages & Disadvantages of Existing Systems

System Type	Advantage	Disadvantages
RFID card system	Simple	Fraudulent usage
Fingerprint system	Accurate	Time-consuming
Voice recognition system	Contactless	Less accurate compared to Others
Iris recognition system	Accurate	Privacy Invasion

3.2 Limitations of Existing Work:

- **Sensitivity to Lighting and Environmental Conditions**

Face recognition systems depend heavily on proper lighting and clear visibility of the face. Poor lighting, shadows, backlight, or low-quality cameras can reduce recognition accuracy. Variations in facial expressions, pose angles, and partial occlusions (such as masks or scarves) may also affect system performance.

- **Limited Real-Time Processing in Earlier Systems**

Many existing systems do not support efficient real-time processing when multiple faces are present simultaneously. As the number of users increases, processing speed may decrease, leading to delays in attendance marking and system response.

- **Lack of Centralized Data Management and Alerts**

Some earlier attendance systems only stored data locally without cloud integration or centralized dashboards. They also lacked real-time notifications to parents or administrators, reducing transparency and remote monitoring capabilities.

- **Scalability Issues**

Several existing implementations were designed for small classrooms or limited users. When deployed in larger institutions, scalability becomes a challenge due to hardware limitations, processing constraints, and database management complexity.

- **High Implementation and Maintenance Cost**

Advanced biometric systems often require high-quality cameras, dedicated servers, and regular maintenance. This increases installation and operational costs, making deployment difficult for smaller institutions.

- **Data Privacy and Security Concerns**

Storing facial data in databases raises concerns regarding data privacy and protection. Without proper encryption and secure access control, sensitive biometric information may be vulnerable to misuse or cyber threats.

3.3 Smart Vision-Based Attendance System:

The proposed system introduces a Smart Vision-Based Attendance System using IoT and Machine Learning to automate and enhance the attendance process. In this system, a camera module connected to an IoT-enabled device such as a Raspberry Pi acts as the primary input unit. The device continuously captures live video streams from the classroom or workplace environment. The captured frames are processed using a Machine Learning-based face recognition algorithm such as Local Binary Pattern Histogram (LBPH), which compares detected faces with pre-stored images in the database to accurately identify individuals and mark attendance automatically.

To enhance reliability and security, the system incorporates real-time data logging and notification features. Once a face is recognized, attendance details are instantly recorded in a centralized database and can be exported in Excel format for institutional use. In case of absence or during predefined time periods, automated SMS or email notifications can be sent to parents or administrators. The system also supports real-time display of recognized names on an LCD or monitor for visual confirmation. By integrating IoT connectivity, database management, and intelligent face recognition, the proposed system ensures contactless operation, improved accuracy, and efficient attendance monitoring in modern educational environments.

3.4 Flow Chart:

In the proposed system, the process begins by storing registered students' facial images in the system database. Each image is saved along with details such as the student's name, roll number, and class. These images serve as reference data for the recognition process.

When the system starts, the camera captures real-time images of students as they enter the classroom. The captured images are continuously processed and compared with the images stored in the database using a face recognition algorithm.

If a match is found, the system identifies the student and marks them as present. The attendance details are recorded, and a file containing the student's information and attendance status is generated. If no match is found, the student is marked absent, and a notification message is sent to the parent.

Finally, the generated attendance file is automatically uploaded to the web server for centralized storage and easy monitoring. This process runs continuously to ensure real-time, contactless, and automated attendance management.

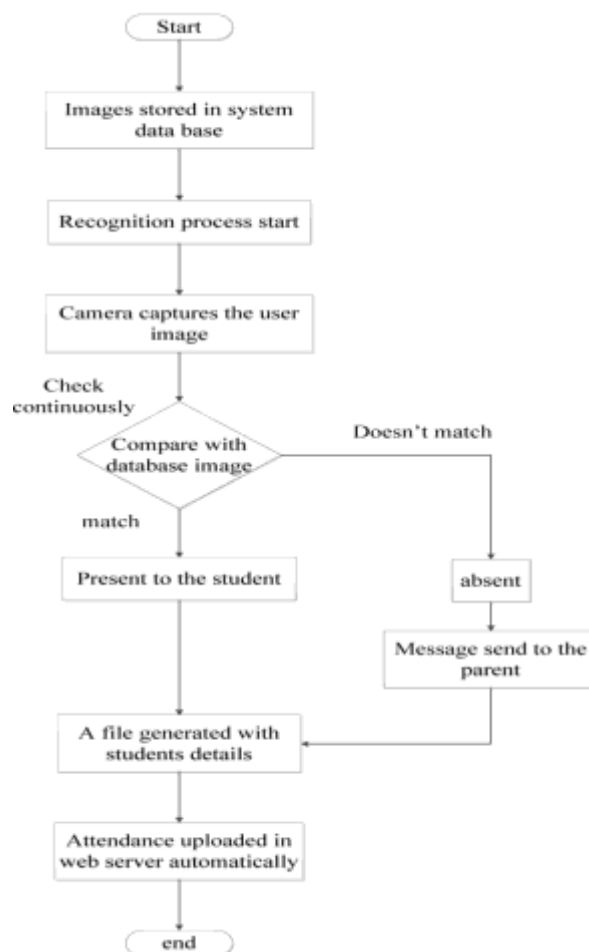


Figure 3.4: Flow Chart of the Smart Vision Based Attendance System

3.5 Working of the Smart Vision-Based Attendance System

The proposed Smart Vision-Based Attendance System is designed to automate attendance marking using face recognition technology integrated with Machine Learning and IoT. The system captures real-time images of students, detects and recognizes faces using the LBPH algorithm, and compares them with stored database records to mark attendance automatically. The recorded data is then uploaded to a cloud platform for centralized storage and monitoring, with optional notification alerts for parents. This approach ensures a contactless, accurate, and efficient attendance management process suitable for modern educational institutions.

3.5.1 IoT in the Smart Vision-Based Attendance System

Internet of Things (IoT) plays an important role in enabling real-time communication and centralized data management in the proposed Smart Vision-Based Attendance System.

After the face recognition process is completed and attendance is marked, the recorded data is automatically transmitted to a cloud server through an internet connection. This transmission is done using IoT-based communication modules integrated with the system.

The IoT component ensures that attendance data is updated instantly on the web platform. This allows administrators and faculty members to access attendance records remotely from any location. The data stored in the cloud is secure, organized and easy to retrieve for report generation and monitoring purposes.

Additionally, IoT enables automatic notification services. When a student is marked absent, the system can send alerts to parents via SMS, mobile applications. This real-time communication improves transparency and strengthens the connection between the institution and parents.

Thus, IoT integration makes the system scalable, efficient, and suitable for modern educational institutions by ensuring seamless data transfer, remote accessibility, and instant notifications.

3.5.2 ML in the Smart Vision-Based Attendance System

Machine Learning (ML) plays a crucial role in enabling intelligent face recognition in the proposed Smart Vision-Based Attendance System. The system uses the Local Binary Pattern Histogram (LBPH) algorithm for identifying and recognizing student faces accurately.

Machine Learning allows the system to learn facial patterns from stored images and make decisions based on extracted features. Instead of simply comparing raw images, the ML

algorithm extracts unique facial characteristics and converts them into numerical feature representations. This improves recognition accuracy and reliability.

The ML process mainly consists of two phases:

1. Training Phase
2. Recognition Phase

LBPH Algorithm Used for Face Recognition

The Local Binary Pattern Histogram (LBPH) algorithm is a texture-based face recognition method. It works by analyzing the local features of the face image.

Step 1: Image Acquisition

The system captures the student's face image and converts it into a grayscale image for processing.

Step 2: Local Binary Pattern (LBP) Operation

- A 3×3 window is selected around each pixel.
- The central pixel value is taken as a threshold.
- Neighboring pixels are compared with the central pixel.
- If the neighbor value is greater than or equal to the threshold, it is assigned '1', otherwise '0'.
- These binary values are combined to form a binary number.
- The binary number is converted into a decimal value.

This process generates a new image that highlights important facial texture features.

The image below **Figure 3.5** shows this procedure:

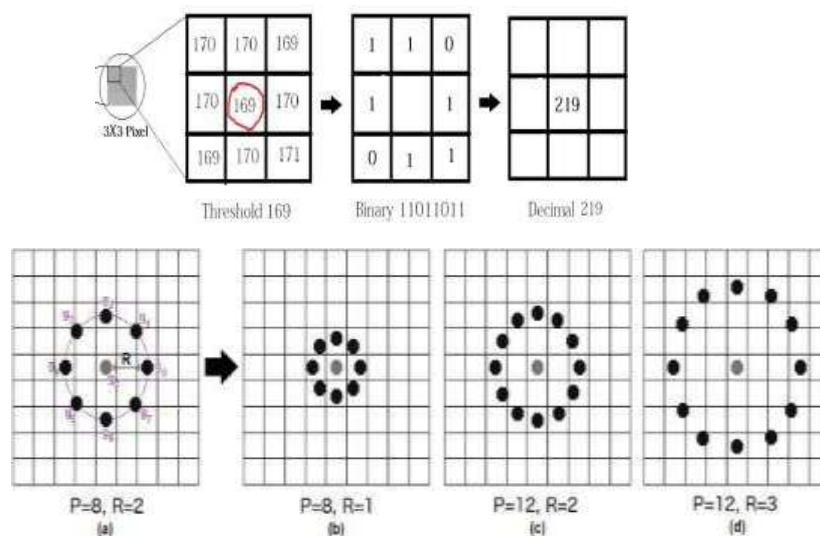


Figure 3.5: LBP Operation

Step 3: Histogram Generation

The processed image is divided into small grids.

For each grid:

- A histogram of pixel intensities (0–255) is calculated.
- All histograms are combined into a single feature vector.

This final histogram represents the facial features of the student.

Step 4: Training Phase

During training, the system stores histograms of all registered students along with their unique IDs (roll numbers).

Step 5: Recognition Phase

When a new face is captured:

- The same LBP and histogram process is applied.
- The generated histogram is compared with stored histograms.
- The Euclidean distance between histograms is calculated.

The student with the minimum distance is identified as the matched person.

If the distance value is below a predefined threshold, the face is recognized and attendance is marked as “Present.” Otherwise, it is marked as unknown.

3.6 Advantages of the Smart Vision-Based Attendance System

The proposed Smart Vision-Based Attendance System offers several advantages over traditional attendance methods:

1. Contactless Operation

The system uses face recognition, eliminating the need for physical contact with devices such as fingerprint scanners or RFID cards.

2. High Accuracy

Machine Learning-based face recognition reduces proxy attendance and improves identification reliability.

3. Time Efficient

Attendance is marked automatically in real time without manual intervention, saving classroom time.

4. Reduced Human Errors

Automated processing eliminates mistakes that occur in manual attendance marking.

5. Real-Time Monitoring

Attendance data is updated instantly and can be monitored remotely.

6. Centralized Cloud Storage

IoT integration allows secure storage and easy access to attendance records from anywhere.

7. Automatic Notifications

Parents or administrators can receive instant alerts regarding student absence.

8. Scalable and Flexible

The system can be implemented in schools, colleges, and other institutions with minimal modification.

CHAPTER 4

HARDWARE IMPLEMENTATION

4.1 Block Diagram:

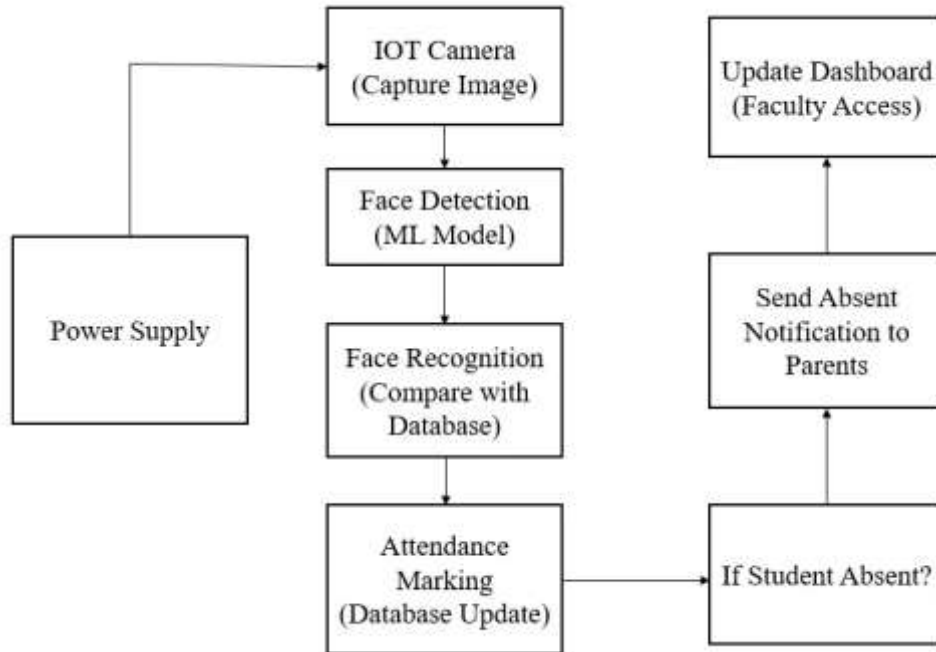


Figure 4.1: Block Diagram of Smart Vision Based Attendance System

The block diagram shown in Figure 5.1 illustrates the working process of the proposed IoT and Machine Learning-based Smart Attendance Management System. The system begins with the Power Supply, which provides the necessary electrical support to all hardware components, including the IoT camera and processing unit.

The IoT Camera (Capture Image) module captures real-time images of students as they enter the classroom or monitored area. These captured images are then forwarded to the Face Detection (ML Model) module, where a machine learning-based algorithm detects the presence of human faces within the frame. This module identifies facial regions from the captured image and prepares them for further processing.

Once faces are detected, the system moves to the Face Recognition (Compare with Database) module. In this stage, the detected facial features are compared with pre-stored images in the database. The recognition algorithm extracts unique facial features and matches them against registered student records to verify identity accurately.

After successful recognition, the Attendance Marking (Database Update) module updates the attendance record in the centralized cloud database. This ensures that attendance is stored securely and can be accessed in real time.

The system then checks the condition “If Student Absent?” to verify whether a registered student has not been detected within the specified time frame.

If a student is marked absent, the system activates the Send Absent Notification to Parents module, which automatically sends alerts or notifications to parents through SMS, email, or mobile application services. Simultaneously, the Update Dashboard (Faculty Access) module updates the web or mobile dashboard, allowing faculty members and administrators to monitor attendance records, view reports, and analyze student presence data in real time.

The overall system ensures automation, real-time processing, secure data management, and improved transparency. By integrating IoT devices, machine learning-based face recognition, cloud storage, and automated notifications, the proposed block diagram represents a scalable and contactless solution for modern attendance management systems.

4.2 Hardware Implementation:

The hardware implementation of the Smart Vision-Based Attendance System is designed to ensure efficient, real-time face recognition and automated attendance tracking with minimal human intervention. The primary hardware components include a camera module, an IoT-enabled processing device, a buzzer for alerts and necessary power and storage units.

The processing unit serves as the core of the system, handling image acquisition, face recognition, attendance marking, real-time notifications and head counting. It features sufficient processing power and memory to run Machine Learning algorithms, such as Local Binary Pattern Histogram (LBPH), in real time. Its connectivity options allow seamless integration with sensors, Wi-Fi modules, buzzers, and cloud services for data storage and dashboard monitoring.

The camera module is selected for its ability to capture high quality images in various lighting conditions, ensuring accurate face recognition in classrooms. The camera continuously streams video frames to the processing unit, where faces are detected, recognized and recorded automatically. The system also counts the total number of individuals present in the classroom helping administrators monitor occupancy and manage attendance efficiently. The buzzer module is incorporated to provide audible alerts after predefined periods, signaling the end of a session or class. The hardware design focuses on scalability, reliability, and energy efficiency, making it suitable for educational institutions and professional environments.



Figure 4.2: Hardware Implementation of Smart Vision Based Attendance System

4.2.1:Raspberry Pi

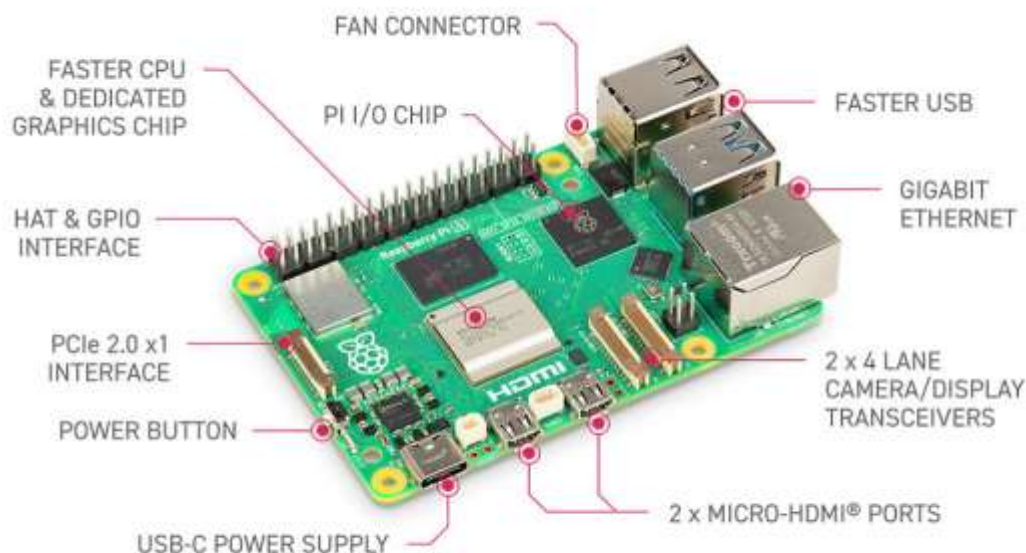


Figure 4.3: Hardware Layout of Raspberry Pi 4 B Model

A brief description of the components on the Pi

1. Processor / SoC (System on Chip)

The Raspberry Pi has a Broadcom BCM2835 System on Chip module, as shown in **Figure 4.3**. It has a ARM1176JZF-S processor. The Broadcom SoC used in the Raspberry Pi is equivalent to a chip used in an old smartphone. While operating at 700 MHz by default, the Raspberry Pi provides a real world performance roughly equivalent to the 0.041 GFLOPS. On the CPU level the performance is similar to a 300 MHz Pentium II of 1997-1999, but the GPU, however, provides 1 Gpixel/s, 1.5 Gtexel/s or 24 GFLOPS of general purpose compute.

and the graphics capabilities of the Raspberry Pi are roughly equivalent to the level of performance of the Xbox of 2001. The Raspberry Pi chip operating at 700 MHz by default, will not become hot enough to need a heatsink or special cooling.

2. Power source

The Pi is a device which consumes 700mA or 3W or power. It is powered by a MicroUSB charger or the GPIO header. Any good smartphone charger will do the work of powering the Pi.

3. SD Card

The Raspberry Pi does not have any onboard storage available. The operating system is loaded on a SD card which is inserted on the SD card slot on the Raspberry Pi. The operating system can be loaded on the card using a card reader on any computer.

4. GPIO

GPIO – General Purpose Input Output General-purpose input/output (GPIO) is a generic pin on an integrated circuit whose behaviour, including whether it is an input or output pin, can be controlled by the user at run time.

GPIO pins have no special purpose defined, and go unused by default. The idea is that sometimes the system designer building a full system that uses the chip might find it useful to have a handful of additional digital control lines, and having these available from the chip can save the hassle of having to arrange additional circuitry to provide them.

GPIO capabilities may include:

- GPIO pins can be configured to be input or output
- GPIO pins can be enabled/disabled
- Input values are readable (typically high=1, low=0)
- Output values are writable/readable
- Input values can often be used as IRQs (typically for wakeup events)

The production Raspberry Pi board has a 26-pin 2.54 mm (100 mil) expansion header, marked as P1, arranged in a 2x13 strip. They provide 8 GPIO pins plus access to I²C, SPI, UART), as well as +3.3 V, +5 V and GND supply lines. Pin one is the pin in the first column and on the bottom row.

As shown in **Figure 4.4**, the GPIO header allows flexible interfacing for sensors, actuators and communication modules, making the Raspberry Pi highly adaptable for embedded system projects.

Pin#	NAME		NAME	Pin#
01	3.3v DC Power		DC Power 5v	02
03	GPIO02 (SDA1 , I ² C)		DC Power 5v	04
05	GPIO03 (SCL1 , I ² C)		Ground	06
07	GPIO04 (GPIO_GCLK)		(TXD0) GPIO14	08
09	Ground		(RXD0) GPIO15	10
11	GPIO17 (GPIO_GEN0)		(GPIO_GEN1) GPIO18	12
13	GPIO27 (GPIO_GEN2)		Ground	14
15	GPIO22 (GPIO_GEN3)		(GPIO_GEN4) GPIO23	16
17	3.3v DC Power		(GPIO_GEN5) GPIO24	18
19	GPIO10 (SPI_MOSI)		Ground	20
21	GPIO09 (SPI_MISO)		(GPIO_GEN6) GPIO25	22
23	GPIO11 (SPI_CLK)		(SPI_CE0_N) GPIO08	24
25	Ground		(SPI_CE1_N) GPIO07	26
27	ID_SD (I ² C ID EEPROM)		(I ² C ID EEPROM) ID_SC	28
29	GPIO05		Ground	30
31	GPIO06		GPIO12	32
33	GPIO13		Ground	34
35	GPIO19		GPIO16	36
37	GPIO26		GPIO20	38
39	Ground		GPIO21	40

Figure 4.4: GPIO Header

5. DSI Connector

Display Serial Interface (DSI) is a specification by the Mobile Industry Processor Interface (MIPI) Alliance aimed at reducing the cost of display controllers in a mobile device. It is commonly targeted at LCD and similar display technologies. It defines a serial bus and a communication protocol between the host (source of the image data) and the device (destination of the image data). A DSI compatible LCD screen can be connected through the DSI connector, although it may require additional drivers to drive the display.

6. RCA Video

RCA Video outputs (PAL and NTSC) are available on all models of Raspberry Pi. Any television or screen with a RCA jack can be connected with the RPi.

As shown in the below **Figure 4.5**, the RCA output allows the Raspberry Pi to interface with standard analog displays for video output.



Figure 4.5: RCA Video Connector

7. Audio Jack

A standard 3.5 mm TRS connector is available on the RPi for stereo audio output. Any headphone or 3.5mm audio cable can be connected directly. Although this jack cannot be used for taking audio input, USB mics or USB sound cards can be used.

8. Status LEDs

There are 5 status LEDs on the RPi that show the status of various activities as follows: “OK” - SD Card Access (via GPIO16) - labelled as "OK" on Model B Rev1.0 boards and "ACT" on Model B Rev2.0 and Model A boards “POWER” - 3.3 V Power - labelled as "PWR" on all boards “FDX” - Full Duplex (LAN) (Model B) - labelled as "FDX" on all boards “LNK” - Link/Activity (LAN) (Model B) - labelled as "LNK" on all boards.

9. USB 2.0 Port

USB 2.0 ports are the means to connect accessories such as mouse or keyboard to the Raspberry Pi. There is 1 port on Model A, 2 on Model B and 4 on Model B+. The number of ports can be increased by using an external powered USB hub which is available as a standard Pi accessory.

10. Ethernet

Ethernet port is available on Model B and B+. It can be connected to a network or internet using a standard LAN cable on the Ethernet port. The Ethernet ports are controlled by Microchip LAN9512 LAN controller chip.

11. CSI connector

CSI-Camera Serial Interface is a serial interface designed by MIPI (Mobile Industry Processor Interface) alliance aimed at interfacing digital cameras with a mobile processor. The RPi foundation provides a camera specially made for the Pi which can be connected with the Pi using the CSI connector.

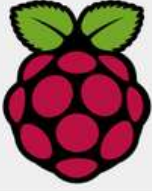
12. JTAG headers

JTAG is an acronym for ‘Joint Test Action Group’, an organisation that started back in the mid 1980's to address test point access issues on PCB with surface mount devices. The organisation devised a method of access to device pins via a serial port that became known as the TAP (Test Access Port). In 1990 the method became a recognised international standard (IEEE Std 1149.1). Many thousands of devices now include this standardised port as a feature to allow test and design engineers to access pins.

13. HDMI

HDMI – High Definition Multimedia Interface HDMI 1.3 a type A port is provided on the RPi to connect with HDMI screens.

Table 4.1: Specifications of Raspberry pi

	Raspberry Pi 3 Model B	Raspberry Pi Zero	Raspberry Pi 2 Model B	Raspberry Pi Model B+
Introduction Date	2/29/2016	11/25/2015	2/2/2015	7/14/2014
SoC	BCM2837	BCM2835	BCM2836	BCM2835
CPU	Quad Cortex A53 @ 1.2GHz	ARM11 @ 1GHz	Quad Cortex A7 @ 900MHz	ARM11 @ 700MHz
Instruction set	ARMv8-A	ARMv6	ARMv7-A	ARMv6
GPU	400MHz VideoCore IV	250MHz VideoCore IV	250MHz VideoCore IV	250MHz VideoCore IV
RAM	1GB SDRAM	512 MB SDRAM	1GB SDRAM	512MB SDRAM
Storage	micro-SD	micro-SD	micro-SD	micro-SD
Ethernet	10/100	none	10/100	10/100
Wireless	802.11n / Bluetooth 4.0	none	none	none
Video Output	HDMI / Composite	HDMI / Composite	HDMI / Composite	HDMI / Composite
Audio Output	HDMI / Headphone	HDMI	HDMI / Headphone	HDMI / Headphone
GPIO	40	40	40	40
Price	\$35	\$5	\$35	\$35

4.2.2: Camera Module

The Camera Module is a crucial component of the proposed Smart Attendance Management System, responsible for capturing real-time images for face detection and recognition. In this project, an IoT-enabled camera module is used to 35 continuously monitor the classroom environment and acquire high-quality facial images of students as they enter or sit in the classroom.

The camera operates by converting light into electronic signals through an image sensor, typically a CMOS (Complementary Metal-Oxide Semiconductor) sensor. When light falls on the sensor, it generates electrical signals proportional to the intensity of the light. These signals are then processed and converted into digital image data, which can be analyzed by machine learning algorithms.

The captured frames are transmitted to the processing unit, such as a Raspberry Pi or microcontroller-based system, where face detection algorithms identify human faces within the image. The quality of recognition largely depends on the resolution, frame rate and lighting conditions supported by the camera module. A higher resolution ensures clearer facial features, improving recognition accuracy, while an appropriate frame rate enables smooth real-time processing.

In low-light or varying illumination conditions, the camera module may include features such as infrared support, auto exposure adjustment and white balance control to maintain image clarity. These features help enhance detection accuracy and reduce false negatives during attendance marking.

The camera module interfaces with the IoT system through USB or CSI (Camera Serial Interface), allowing seamless communication with the processing device. The captured image data is then forwarded to the Face Detection and Face Recognition modules for further processing.

Overall, the camera module serves as the primary input device of the system, enabling real-time image acquisition, accurate face recognition and automated attendance marking. Its performance directly impacts system reliability, accuracy and efficiency, as shown in the below **Figure 4.6**.



Figure 4.6: Camera Module

4.2.3 LCD (Liquid Crystal Display)

A liquid crystal display (LCD) is a thin, flat display device made up of any number of color or monochrome pixels arrayed in front of a light source or reflector. Each pixel consists of a column of liquid crystal molecules suspended between two transparent electrodes and two polarizing filters, the axes of polarity of which are perpendicular to each other. Without the liquid crystals between them, light passing through one would be blocked by the other. The liquid crystal twists the polarization of light entering one filter to allow it to pass through the other.

A program must interact with the outside world using input and output devices that communicate directly with a human being. One of the most common devices attached to an controller is an LCD display. Some of the most common LCDs connected to the controllers are 16X1, 16x2 and 20x2 displays. This means 16 characters per line by 1 line 16 characters per line by 2 lines and 20 characters per line by 2 lines, respectively.

Many microcontroller devices use 'smart LCD' displays to output visual information. LCD displays designed around LCD NT-C1611 module, are inexpensive, easy to use, and it is even possible to produce a readout using the 5X7 dots plus cursor of the display. They have a standard ASCII set of characters and mathematical symbols. For an 8-bit data bus, the display requires a +5V supply plus 10 I/O lines (RS RW D7 D6 D5 D4 D3 D2 D1 D0). For a 4-bit data bus it only requires the supply lines plus 6 extra lines (RS RW D7 D6 D5 D4). When the LCD display is not enabled, data lines are tri-state and they do not interfere with the operation of the microcontroller.

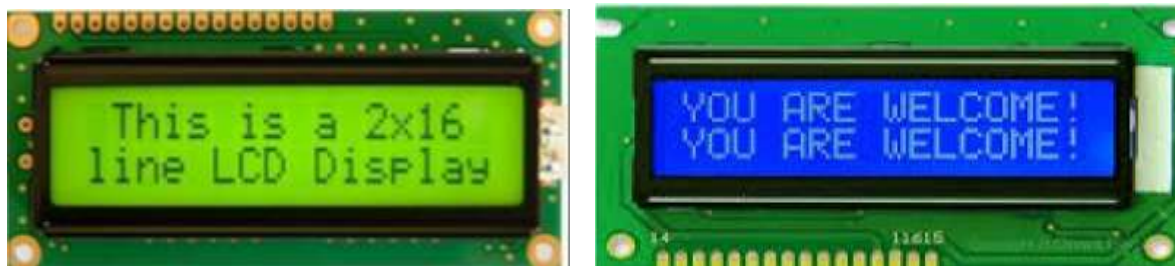
Features:

- Interface with either 4-bit or 8-bit microprocessor
- Display data RAM
- 80x8 bits (80 characters)
- Character generator ROM
- 160 different 5×7 dot-matrix character patterns.
- Character generator RAM
- 8 different user programmed 5×7 dot-matrix patterns.
- Display data RAM and character generator RAM may be Accessed by the microprocessor
- Numerous instructions
- Clear Display, Cursor Home, Display ON/OFF, Cursor ON/OFF, Blink Character, Cursor Shift, Display Shift.
- Built-in reset circuit is triggered
- Built-in oscillator.

Data can be placed at any location on the LCD. For 16x1 LCD, the address locations are:

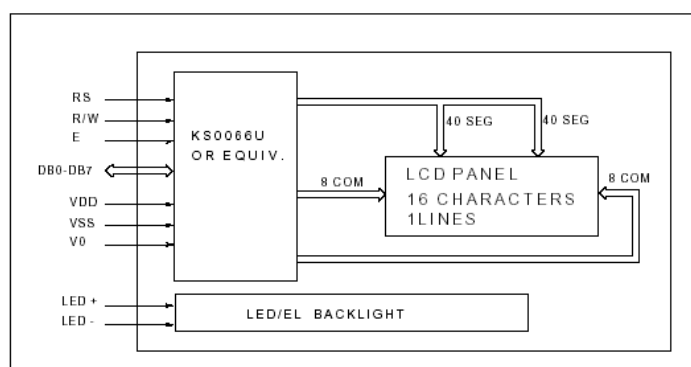
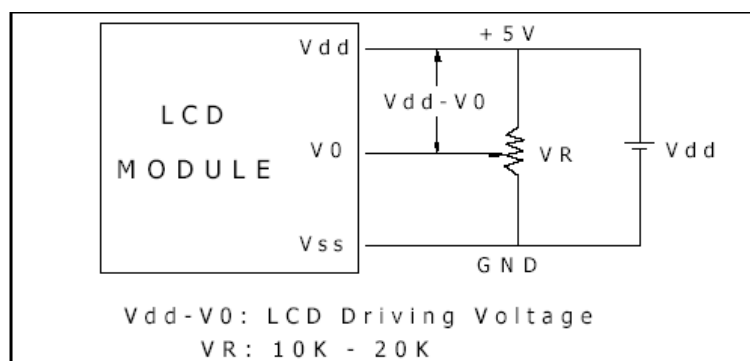
Table 4.2: Address locations for a 1x16 line LCD

POSITION		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
ADDRESS	LINE1	00	01	02	03	04	05	06	07	40	41	42	43	44	45	46	47

Shapes and sizes:**Figure 4.7: LCD Display Shape and Size**

Even limited to character based modules, there is still a wide variety of shapes and sizes available. Line lengths of 8,16,20,24,32 and 40 characters are all standard, in one, two and four line versions.

Several different LC technologies exist. “supertwist” types, for example, offer improved contrast and viewing angle over the older “twisted nematic” types. Some modules are available with back lighting, so so that they can be viewed in dimly-lit conditions. The back lighting may be either “electro-luminescent”, requiring a high voltage inverter circuit, or simple LED illumination, as shown in the above **Figure 4.7** and below **Figure 4.8,4.9**.

Electrical blockdiagram:**Figure 4.8:Block Diagram Of Electrical****Power supply for lcd driving:****Figure 4.9: Power Supply For LCD Driving**

Pin Description:

Most LCDs with 1 controller has 14 Pins and LCDs with 2 controller has 16 Pins (two pins are extra in both for back-light LED connections),as shown in below **Figure 4.10**.

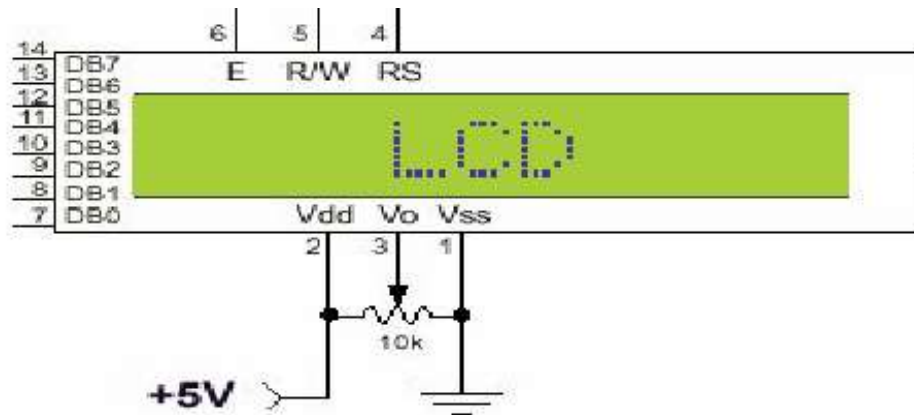


Figure 4.10: Pin diagram of 1x16 lines lcd

Table 4.3: Pin Description of LCD Display

PIN	SYMBOL	FUNCTION
1	Vss	Power Supply(GND)
2	Vdd	Power Supply(+5V)
3	Vo	Contrast Adjust
4	RS	Instruction/Data Register Select
5	R/W	Data Bus Line
6	E	Enable Signal
7-14	DB0-DB7	Data Bus Line
15	A	Power Supply for LED B/L(+)
16	K	Power Supply for LED B/L(-)

Control Lines:

- **EN:**

Line is called "Enable." This control line is used to tell the LCD that you are sending it data. To send data to the LCD, your program should make sure this line is low (0) and then set the other two control lines and/or put data on the data bus. When the other lines are completely ready, bring EN high (1) and wait for the minimum amount of time required by the LCD datasheet (this varies from LCD to LCD), and end by bringing it low (0) again.

- **RS:**

Line is the "Register Select" line. When RS is low (0), the data is to be treated as a command or special instruction (such as clear screen, position cursor, etc.). When RS is high (1), the data being sent is text data which could be displayed on the screen. For example, to display the letter "T" on the screen you would set RS high.

- **RW:**

Line is the "Read/Write" control line. When RW is low (0), the information on the data bus is being written to the LCD. When RW is high (1), the program is effectively querying (or reading) the LCD. Only one instruction ("Get LCD status") is a read command. All others are write commands, so RW will almost always be low. Finally, the data bus consists of 4 or 8 lines (depending on the mode of operation selected by the user). In the case of an 8-bit data bus, the lines are referred to as DB0, DB1, DB2, DB3, DB4, DB5, DB6, and DB7.

Logic status on control lines:

- E - 0 Access to LCD disabled - 1 Access to LCD enabled
- R/W - 0 Writing data to LCD - 1 Reading data from LCD
- RS - 0 Instructions - 1 Character

Writing data to the LCD:

- 1) Set R/W bit to low
- 2) Set RS bit to logic 0 or 1 (instruction or character)
- 3) Set data to data lines (if it is writing)
- 4) Set E line to high
- 5) Set E line to low

Read data from data lines (if it is reading)on LCD:

- 1) Set R/W bit to high
- 2) Set RS bit to logic 0 or 1 (instruction or character)
- 3) Set data to data lines (if it is writing)
- 4) Set E line to high
- 5) Set E line to low

Entering Text:

Text can be entered into the display module using hexadecimal codes instead of binary for easier operation. The RS pin must be set high (logic 1) to send character data, after which the desired character code is applied to the data pins. The E (Enable) switch is then triggered to display the character and this process is repeated to enter additional text.

4.2.4 GSM (Global System for Mobile communications)



Figure 4.11: GSM Module

The GSM (Global System for Mobile Communication) module is an essential communication component used in the proposed Smart Attendance Management System to send notifications and alerts to parents or guardians. It enables wireless communication over cellular networks, allowing the system to transmit SMS messages in real time whenever a student is marked absent.

As shown in **Figure 4.11**, The GSM module operates using a SIM card registered with a mobile network provider. It communicates with the processing unit, such as a Raspberry Pi or microcontroller, through serial communication (UART protocol). When the system detects that a student is absent, a command is sent to the GSM module using AT commands, which instructs the module to send an SMS notification to the registered phone number.

The module functions on standard GSM frequency bands (such as 900 MHz or 1800 MHz), ensuring reliable communication over long distances. Unlike Wi-Fi-based notifications, GSM communication does not depend on internet connectivity, making it suitable for areas with limited network infrastructure.

In this project, the GSM module enhances automation by instantly informing parents about student absence, improving transparency and safety. It eliminates the need for manual communication by faculty and ensures that attendance information is delivered quickly and efficiently.

The advantages of using a GSM module include wide network coverage, low implementation cost, ease of integration with microcontrollers, and reliable SMS delivery. However, it requires a valid SIM card, sufficient signal strength, and incurs minimal messaging charges based on the service provider.

Overall, the GSM module plays a vital role in enabling real-time alert mechanisms in the smart attendance system, ensuring effective communication between the institution and parents.

4.2.5 Buzzer Module



Figure 4.12: Buzzer Module

The buzzer is an audio signaling device used in the proposed Smart Attendance Management System to provide alert notifications at the end of every class session. It serves as an audible indicator to notify students and faculty that the class duration has been completed and attendance processing has been finalized.

In this system, the buzzer is connected to the microcontroller or processing unit. Once the predefined class time is reached, the controller automatically triggers the buzzer by sending a voltage signal. Upon receiving the signal, the buzzer converts electrical energy into sound energy and produces a beep sound to indicate the completion of the session.

As shown in **Figure 4.12**, The buzzer used in the system may be either an active or passive type. An active buzzer generates sound immediately when powered, while a passive buzzer requires a frequency signal from the controller. Due to its simplicity and low power consumption, the active buzzer is commonly preferred for alert applications.

By providing a clear audio signal after every class, the buzzer ensures timely session transitions and confirms that attendance has been successfully recorded and updated in the database. It enhances system automation by eliminating the need for manual reminders or bell systems.

Overall, the buzzer module functions as an automated alert mechanism, improving classroom time management and ensuring smooth operation of the smart attendance system.

CHAPTER 5

SOFTWARE IMPLEMENTATION

5.1 RealVNC Viewer:

The RealVNC Viewer is a remote desktop application that enables users to access and control computers or embedded systems, such as a Raspberry Pi, from anywhere. It is part of the RealVNC Connect software suite, providing secure and encrypted remote access solutions. With RealVNC Viewer, users can connect to a device running RealVNC Server, allowing them to interact with the graphical user interface as if they were physically present. This software is widely used for remote administration, troubleshooting, and monitoring, making it ideal for IoT-based attendance systems. It supports multiple operating systems, including Windows, macOS, Linux, and Raspberry Pi OS, ensuring cross-platform compatibility. Additionally, RealVNC uses strong encryption protocols to protect data transmission, making it a secure choice for applications where privacy and security are crucial.

In the context of the Smart Vision-Based Attendance System, RealVNC Viewer plays a key role in remote monitoring and management. Since the system runs on a Raspberry Pi or IoT-enabled device, users can access it remotely via RealVNC Viewer to configure settings, monitor real-time video feeds, and troubleshoot issues without needing physical access. This feature is particularly useful for administrators or teachers who need to manage attendance systems in multiple classrooms. Moreover, RealVNC allows file transfers, clipboard sharing, and multi-user collaboration, enhancing the efficiency of system maintenance and data retrieval.

By integrating RealVNC Viewer with the Smart Vision-Based Attendance System, users can ensure seamless operation, quick troubleshooting, and efficient remote management, reducing manual intervention while maintaining secure and accurate attendance tracking.

5.2 How to Install RealVNC Viewer:

Let's see how to install RealVNC Viewer:

1.Download RealVNC Viewer:

- **Visit official website:** <https://www.realvnc.com/en/connect/download/viewer/>
- **Select the appropriate version for Windows or macOS.**

2.Install the Software:

Open the downloaded file and follow the installation instructions.

- For Windows, double-click the .exe file and proceed with the installation.
- For macOS, open the .dmg file and drag the VNC Viewer icon into the Applications folder.

3.Run RealVNC Viewer:

Launch the application and log in with your RealVNC account (if required)

- Enter the IP address of the remote device running RealVNC Server to establish a connection.
- Open the downloaded file and follow the installation instructions.

5.3 Implementation in RealVNC Viewer:

5.3.1 Opening RealVNC Viewer:

RealVNC Viewer is a remote desktop application that allows users to connect to and control Raspberry Pi 4 from another device.

To open RealVNC Viewer:

- Launch the application from the Start menu (Windows), Applications folder (macOS), or by running `vncviewer` in the terminal (Linux).
- Enter the Raspberry Pi's IP address or hostname in the VNC Viewer address bar.
- Authenticate using the Raspberry Pi's username and password to establish a secure remote connection.

As shown in the below Figure 5.1, RealVNC Viewer provides a simple and intuitive interface for remotely accessing the Raspberry Pi.

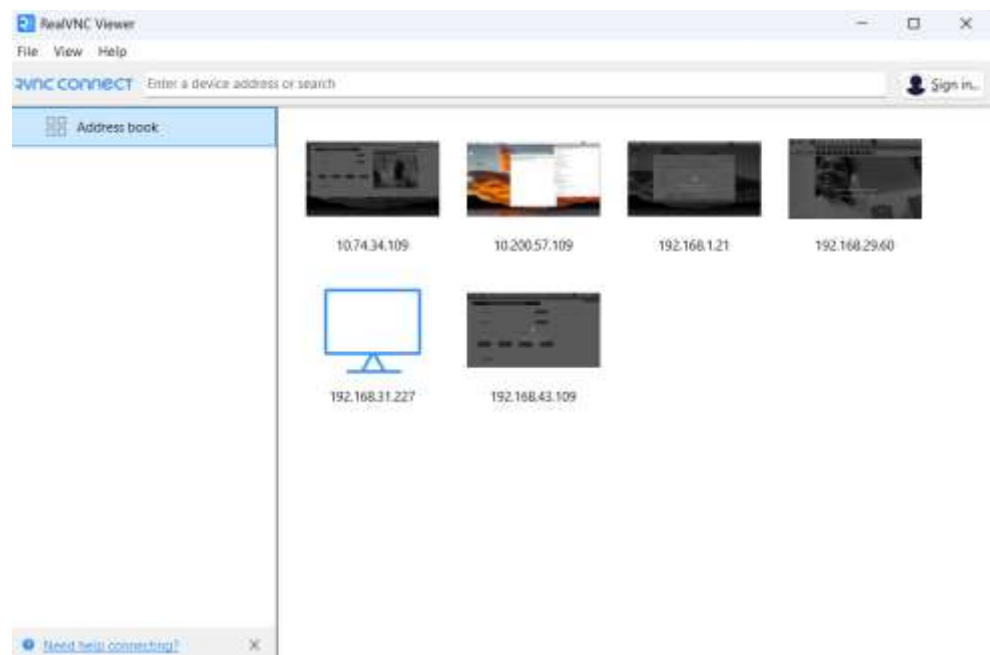


Figure 5.1: RealVNC Viewer

5.3.2 Code Implementation:

All our code is written in Python language. First here is our project directory structure and files.

Smart Vision Based Attendance System

```
|__[Attendance]
|__[Images Unknown]
|__[Student Details]
|__[Training Image]
|__[Training Label]
|__main.py
|__automail.py
|__CaptureImage.py
|__check_camera.py
|__recognize.py
```

All those file in the project directory.

Note: The names inside square brackets ["folder name"] indicate it is a folder.

[Attendance] => It contains all the attendance sheets saved after taking attendance.
[Images Unknown] => Unknown images are placed inside this folder to avoid false positives.
[Student Details] => Here we place Studentdetails.csv file to use while recognizing faces.
[Training image] => After capture dataset of a student, all his/her images are stored here.

5.3.3 Program Execution:

- After establishing the remote connection using RealVNC Viewer, the project directory is opened in the Raspberry Pi terminal. The Python program is executed using the appropriate command. Once the program runs, the camera module is activated automatically.
- The system begins capturing real-time video frames and performs face detection using OpenCV. When a registered student's face is detected, the LBPH algorithm recognizes the face and displays the student's name along with the confidence level on the screen.
- If the face matches the stored database, attendance is marked automatically and saved in the attendance file. If the face is not recognized, the system labels it as "Unknown." The attendance data is then uploaded to the cloud server through IoT integration. Finally, for students marked as absent, an automatic notification message is sent to their parents to inform them about the absence, ensuring timely communication and improved monitoring.

CHAPTER 6

RESULTS

6.1 User Interface of the Smart Vision-Based Attendance System

The user interface (UI) of the Smart Vision-Based Attendance System is designed to be simple, interactive, and user-friendly, allowing administrators or teachers to manage attendance efficiently. The main interface contains the following options:

1. Enter User ID:

- This field allows the administrator to input a unique ID for each student.
- The ID serves as the primary identifier for attendance tracking and database records.

2. Enter Name:

- Alongside the User ID, the administrator can enter the student's name.
- This ensures that the captured face is linked to the correct individual for accurate recognition.

3. Take Images:

- Clicking this option enables the system to capture multiple facial images of the student using the connected camera.
- The images are stored locally for training the face recognition model.

4. Train Images:

- This button triggers the Machine Learning model training process.
- The system uses the captured images to learn and recognize each student's face efficiently.

5. Track Images:

- After training, this feature allows the system to recognize faces in real time.
- It marks attendance automatically, updates the head count, and sends notifications for absent students.

6. Quit:

- This option closes the application safely, ensuring that all data and logs are properly saved.

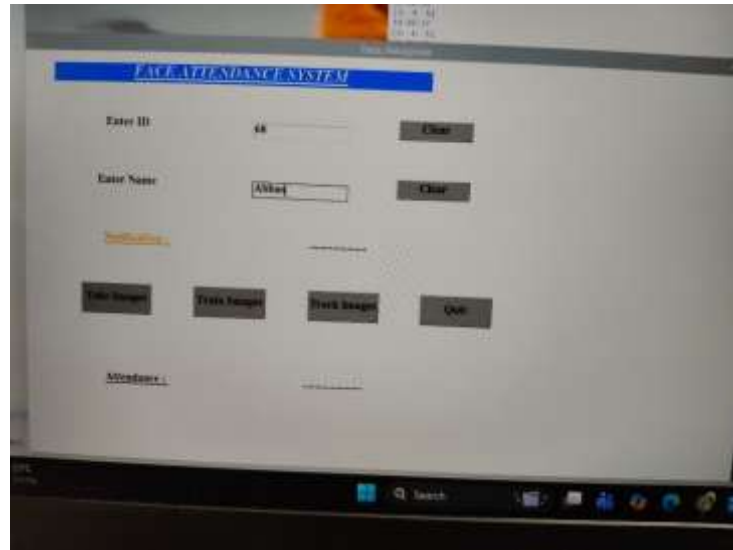


Figure 6.1: User Interface of the Attendance System

6.2 Face Detection and Recognition

The system captures frames from the camera and detects faces in real time. Detected faces are compared with pre-registered images in the database using the LBPH (Local Binary Pattern Histogram) algorithm. When a match is found, the student's name and ID are displayed, and attendance is marked automatically. The system works efficiently for multiple students entering at the same time and under varied lighting conditions, ensuring accurate recognition.

Below, images of registered users and recognized users in real time can be added to demonstrate the system's functionality as shown in **Figure 6.2, 6.3, 6.4, 6.5, and 6.6.**



Figure 6.2: Registered and Recognized User 1

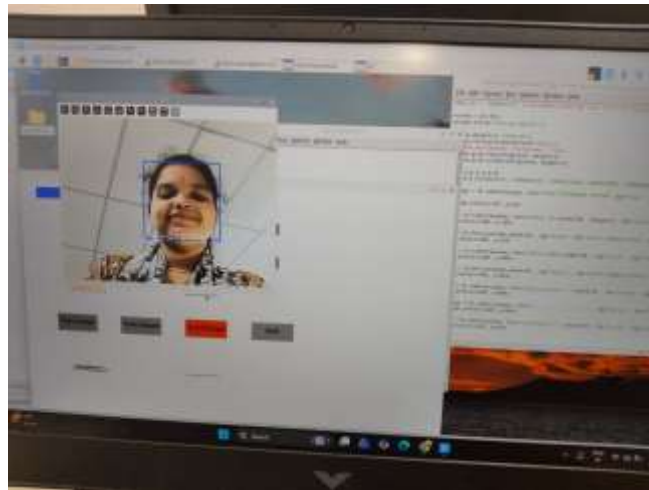


Figure 6.3: Registered and Recognized User 2

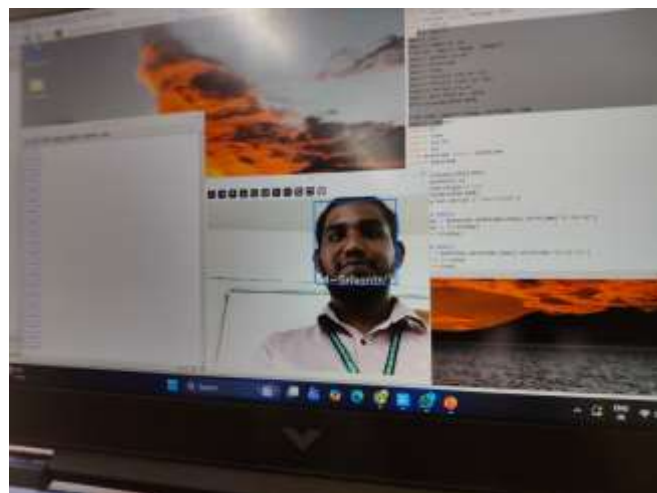


Figure 6.4: Registered and Recognized User 3

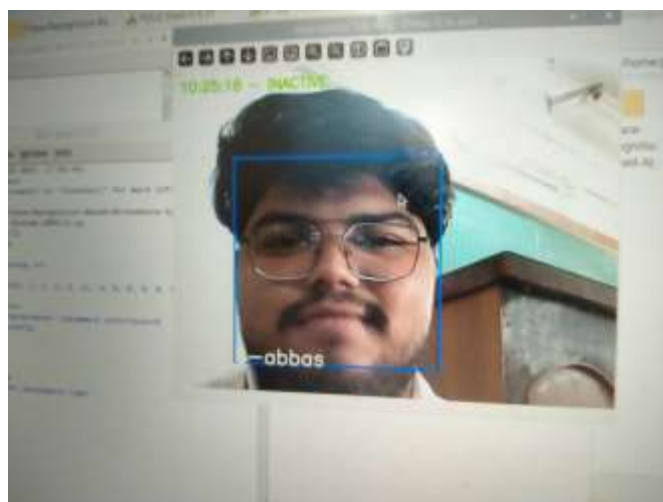


Figure 6.5: Registered and Recognized User 4

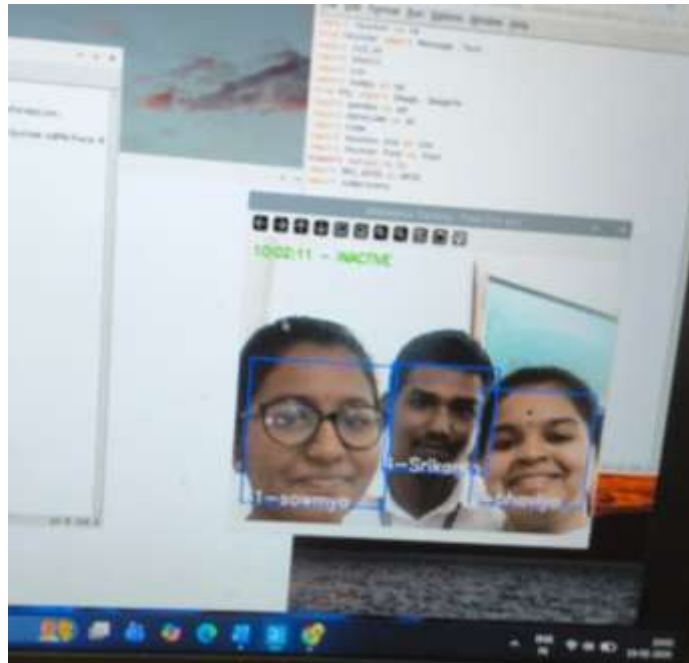


Figure 6.6: Real-Time Recognition of Multiple Registered Users in a Single Frame

6.3 Head Count:

The system continuously monitors the classroom and counts the total number of students present in real time. The head count feature helps teachers verify the number of students in the classroom, ensuring accurate attendance tracking. This count is automatically updated on the user interface alongside recognized faces, providing instant visibility of student presence recognition, as shown in below **Figure 6.7**.



Figure 6.7: Head Count of the Registered Users

6.4 Excel-Based Attendance Log Shared Through Telegram Bot

The system automatically generates an Excel attendance log containing Student ID, Name, Date, Time, and Attendance Status. This log is shared directly with faculty via a Telegram bot, providing real time updates and eliminating the need for manual entry. Faculty can view, download, and archive the attendance records easily, ensuring accuracy and timely management, as shown in below **Figure 6.8**.

	A	B	C	D	E
1	Id	Name	Status	Time	Date
2		1 sowmya	Present	17:47:16	17-02-2026
3		2 Bhaviya	Present	17:47:17	17-02-2026
4		3 Abbas	Absent	N/A	17-02-2026
5		4 Srikanth	Present	17:47:18	17-02-2026
6					

Figure 6.8: Automated Attendance Log Stored in Excel Format

6.5 Real-Time Notifications Sent to Parents

The system sends instant SMS alerts to parents whenever a student is absent. These notifications ensure timely communication and accountability, keeping parents informed in real time while reducing the need for manual follow-up by teachers or administrators. as shown in below **Figure 6.9**.



Figure 6.9: Real-Time SMS Notification Sent to Parents

CONCLUSION AND FUTURE SCOPE

Conclusion:

The Smart Vision-Based Attendance System successfully demonstrates an automated, real-time, and contactless approach to attendance management using IoT and Machine Learning-based face recognition. The system efficiently detects and recognizes students, maintains accurate attendance logs, counts the number of students present, triggers buzzer alerts after periods, and sends instant notifications to parents or faculty via Telegram or SMS. By replacing traditional manual or RFID-based methods, this project enhances accuracy, efficiency, and accountability in educational institutions. The integration of cloud-based storage and dashboard monitoring ensures secure and easily accessible attendance records, while the use of RealVNC Viewer allows remote management and troubleshooting. Overall, the project provides a scalable, user-friendly, and reliable solution suitable for modern classrooms and professional environments.

Future Scope:

The Smart Vision-Based Attendance System can be further enhanced by developing a mobile application for teachers, students, and parents, enabling them to access attendance records and notifications directly from their smartphones. This will make the system more convenient and accessible. The accuracy and robustness of the system can be improved by implementing advanced deep learning-based face recognition algorithms, which can handle larger classrooms, crowded environments, and varying lighting conditions more effectively.

The system can also be extended to monitor multiple classrooms simultaneously using a centralized dashboard, allowing administrators to manage attendance across an entire institution efficiently. For higher security, multi-factor biometric attendance can be incorporated by combining face recognition with other methods, such as fingerprints or RFID, to prevent misuse and ensure authentic attendance.

In addition, integrating analytics and reporting features can help generate detailed attendance trends, monthly summaries, and insights into student behavior, assisting faculty in academic planning. Finally, connecting the system with school management or ERP software can automate grade tracking, performance analysis, and administrative workflows, making the system a fully comprehensive attendance management solution.

BIBLIOGRAPHY

- [1] A. Kumar, S. Mishra, and P. Sharma, "IoT-enabled real-time face recognition attendance system using lightweight machine learning techniques," *International Journal of Intelligent Systems and Applications*, vol. 14, no. 6, pp. 45–56, 2022.
- [2] M. Naveen Raj and R. Vadivel, "Face recognition based attendance system using machine learning with location identification," *World J. Adv. Res. Rev.*, vol. 18, no. 1, pp. 1029–1035, 2023.
- [3] G. Ramani et al., "Smart attendance system using IoT," in *Proc. IEEE Int. Conf. on Communication and Computational Systems (ICCCS)*, 2023.
- [4] N. Ramakrishnan, A. Ajil, A. Ali, A. M. Sundaram, and V. Sanju, "Wi-Fi-based smart attendance monitoring system," in *Proc. IEEE Conf. on Communication Systems, Computing and IT Applications (CSITSS)*, 2023.
- [5] A. R. Ansari and L. Ahmad, "Smart attendance monitoring system using deep neural networks and edge computing," in *Proc. IEEE Int. Conf. on Smart Systems and IoT*, pp. 112–116, Jan. 2023.
- [6] Preetham B R, Prajwal R, and Sanjay K, "Smart Attendance System Using RFID and Face Recognition Technology" 2023.
- [7] K. S. Reddy, P. Kumar, and M. S. Rao, "IoT-based real time face recognition attendance system using classical machine learning techniques," *International Journal of Engineering and Advanced Technology*, vol. 13, no. 2, pp. 245–252, 2023.
- [8] M. R. Islam and M. S. Ahmed, "Vision-based real-time attendance system with weather-resistant cameras for outdoor use," *IEEE Sensors Journal*, vol. 24, no. 14, pp. 13012–13022, Jul. 2024.
- [9] N. Thaleeparambil, A. Biju, and B. Prathap, "Integrated automated attendance system with RFID, Wi-Fi, and visual recognition technology for enhanced classroom security and precise monitoring," in *Proc. IEEE Int. Conf. on Contemporary Computing and Communication (InC4)*, 2024.
- [10] T. Abderraouf, A. A. Wassim, and S. Larabi, "An embedded intelligent system for monitoring," *arXiv preprint*, Jun. 2024.
- [11] S. Das and A. Mukherjee, "Smart classroom attendance management using AI and IoT with real time alerts," *Int. J. Eng. Res. Technol.*, vol. 15, no. 9, pp. 1182–1190, Sep. 2024.
- [12] D. V. Darshan et al., "IoT-based attendance monitoring with facial recognition," in *Proc. IEEE Int. Conf. on Advanced Information Technology (ICAIT)*, 2024.
- [13] A. Raj et al., "Facial recognition-based student attendance system," in *Proc. IEEE Int. Conf. on Recent Innovations in (ICRITO)*, 2024. Technology and Optimization.

- [14] O. Kale, D. Kharat, S. Babar, and C. Barse, “Real-time attendance system using facial recognition and web based integration,” *Int. J. Creative Res. Thoughts*, vol. 13, no. 3, pp. 1–8, Mar. 2025.
- [15] D. K. B., D. C. N., R. H. V., S. N., and T. P. M., “Real time student face recognition attendance system using AI,” *Int. Adv. Res. J. Sci. Eng. Technol.*, vol. 12, no. 5, pp. 1067–1072, May 2025.

PROJECT OUTCOMES MAPPED WITH PROGRAMME SPECIFIC OUTCOMES (PSOs) AND PROGRAMME OUTCOMES (POs)

Course Title: B.Tech Capstone Project

Course Code: IN3535

Department: Internet of Things

Batch Number: B12

Title of the Project: Smart Vision-Based Attendance System Using IoT and ML With Real Time Notifications

Academic Year: 2025 – 2026

1. Project Course Outcomes (COs)

CO No	Course Outcome Statement
CO1	Apply knowledge to design and implement real time solutions for practical Engineering Problems.
CO2	Critically review literature to identify gaps and suggest improvements.
CO3	Conduct research by designing experiments, analyzing data and validating IoT based solutions.
CO4	Design and develop an IoT product considering performance, cost and societal impact.
CO5	Apply ethical principles and communication skills to complete engineering project collaboratively.

2. Program Outcomes (POs)

PO No	Program Outcome Statement
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3	Design / Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5	Engineering tool usage: Create, select and apply appropriate techniques, resources and modern engineering & 17 tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	The Engineer andThe World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	Individual & Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

3. Program Specific Outcomes (PSOs)

PSO No	PSO Statement
PSO1	Implementing innovative, costeffective processes for producing energy efficient and eco-friendly IoT applications.
PSO2	Applying the knowledge of Core, Disciplinary and Interdisciplinary elective courses of Electronics, Communications and Computer Engineering to achieve a logical solution to real world problems

4. CO – PO / PSO Mapping Table (3-High, 2- Moderate, 1- Low)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1	-	-	-	-	-	-	-	-	2	1
CO2	2	2	3	2	3	-	-	-	1	-	-	3	2
CO3	2	2	2	3	3	-	-	-	1	-	-	3	3
CO4	2	3	2	3	2	-	-	-	1	-	-	2	3
CO5	-	-	1	1	2	-	-	2	3	3	3	1	2

5. Sustainable Development Goals (SDG) Alignment

SDG No	Goal	Relevance to the Project
3	Good Health and Well-being	By minimizing manual attendance processes and reducing crowding, the system promotes hygiene and helps prevent the spread of infections, especially in schools and workplaces.
4	Quality Education	Ensures accurate attendance tracking, improves discipline and enables better monitoring of student participation, contributing to enhanced learning outcomes.
9	Industry, Innovation and Infrastructure	The project uses IoT, computer vision, and machine learning to modernize traditional attendance systems, improving efficiency, automation and technological advancement.
11	Sustainable Cities and Communities	Supports smart infrastructure in educational institutions and workplaces, enabling safer, more organized, and technology-driven environments.
16	Peace, Justice and Strong Institutions	Ensures transparency and fairness by eliminating proxy attendance and manual errors, strengthening trust and accountability in institutional systems.

Justification of CO - PO/PSO Mapping (Table 5):

CO	Justification
CO1	Understands fundamentals of IoT, Machine Learning and computer vision to identify real-world problems like manual attendance inefficiencies. Supports PO1 and PO2 through application of engineering knowledge and problem analysis. Contributes to PSO1 by building domain knowledge in smart systems.
CO2	Focuses on design and development of an automated attendance system using IoT devices, cameras and ML algorithms. Strongly maps to PO3 and PO5, while supporting PO2 and PO4. Enhances practical implementation skills aligned with PSO1 and PSO2.
CO3	Involves image capture, face recognition processing and cloud/database integration for storing attendance records. Relates to PO4 and PO5, with support from PO1 and PO2. Develops implementation and system integration skills contributing to PSO1 and PSO2.
CO4	Emphasizes analysis and interpretation of attendance data and real-time notifications for decision-making. Supports PO2 and PO4, while contributing to PO3. Strengthens analytical and problem-solving ability aligned with PSO2.
CO5	Highlights teamwork, communication, and project management during system development and deployment. Strongly maps to PO9, PO10, and PO11, while supporting PO8 and PO5. Encourages collaborative development aligned with PSO1 and PSO2.

Signature(s) of Students:

Signature of Project Supervisor:

1.

2.

3.

4.

APPENDIX A

A.1: Code

```
import tkinter as tk
from tkinter import Message ,Text
import cv2,os
import shutil
import csv
import numpy as np
from PIL import Image, ImageTk
import pandas as pd
import datetime
import time
import tkinter.ttk as ttk
import tkinter.font as font
#import serial_rx_tx
import RPi.GPIO as GPIO
GPIO.setmode(GPIO.BCM)

from time import sleep, strftime, time
import _thread
import os
import time
import serial
import sys
from datetime import datetime
import datetime
buz=2
GPIO.setup(buz,GPIO.OUT)
GPIO.output(buz,1)
GPIO.setwarnings(False)
GPIO.setmode(GPIO.BCM)
ser = serial.Serial ("/dev/ttyS0")
```



```
def date_now():
    today = datetime.datetime.now().strftime("%Y-%m-%d")
    today = str(today)
    return(today)

def time_now():
    now = datetime.datetime.now().strftime("%H:%M:%S")
    now = str(now)
    return(now)

a1_h=16
a1_m=5

a2_h=16
a2_m=7

a3_h=16
a3_m=7

LCD_RS = 26
LCD_E = 19
LCD_D4 = 13
LCD_D5 = 6
LCD_D6 = 21
LCD_D7 = 20

GPIO.setup(LCD_E, GPIO.OUT) # E
GPIO.setup(LCD_RS, GPIO.OUT) # RS
GPIO.setup(LCD_D4, GPIO.OUT) # DB4
GPIO.setup(LCD_D5, GPIO.OUT) # DB5
GPIO.setup(LCD_D6, GPIO.OUT) # DB6
GPIO.setup(LCD_D7, GPIO.OUT) # DB7
```

```
# Define some device constants
```

```
LCD_WIDTH = 16 # Maximum characters per line
LCD_CHR = True
LCD_CMD = False

LCD_LINE_1 = 0x80 # LCD RAM address for the 1st line
LCD_LINE_2 = 0xC0 # LCD RAM address for the 2nd line

# Timing constants
E_PULSE = 0.0005
E_DELAY = 0.0005

def lcd_init():
    # Initialise display
    lcd_byte(0x33,LCD_CMD) # 110011 Initialise
    lcd_byte(0x32,LCD_CMD) # 110010 Initialise
    lcd_byte(0x06,LCD_CMD) # 000110 Cursor move direction
    lcd_byte(0x0C,LCD_CMD) # 001100 Display On,Cursor Off, Blink Off
    lcd_byte(0x28,LCD_CMD) # 101000 Data length, number of lines, font size
    lcd_byte(0x01,LCD_CMD) # 000001 Clear display
    time.sleep(E_DELAY)

def lcd_byte(bits, mode):
    # Send byte to data pins
    # bits = data
    # mode = True for character
    #      False for command

    GPIO.output(LCD_RS, mode) # RS

    # High bits
    GPIO.output(LCD_D4, False)
    GPIO.output(LCD_D5, False)
    GPIO.output(LCD_D6, False)
    GPIO.output(LCD_D7, False)
```

```
if bits&0x10==0x10:
    GPIO.output(LCD_D4, True)
if bits&0x20==0x20:
    GPIO.output(LCD_D5, True)
if bits&0x40==0x40:
    GPIO.output(LCD_D6, True)
if bits&0x80==0x80:
    GPIO.output(LCD_D7, True)
```

```
# Toggle 'Enable' pin
lcd_toggle_enable()
```

```
# Low bits
GPIO.output(LCD_D4, False)
GPIO.output(LCD_D5, False)
GPIO.output(LCD_D6, False)
GPIO.output(LCD_D7, False)
if bits&0x01==0x01:
    GPIO.output(LCD_D4, True)
if bits&0x02==0x02:
    GPIO.output(LCD_D5, True)
if bits&0x04==0x04:
    GPIO.output(LCD_D6, True)
if bits&0x08==0x08:
    GPIO.output(LCD_D7, True)
```

```
# Toggle 'Enable' pin
lcd_toggle_enable()
```

```
def lcd_toggle_enable():
    # Toggle enable
    time.sleep(E_DELAY)
    GPIO.output(LCD_E, True)
    time.sleep(E_PULSE)
```

```
GPIO.output(LCD_E, False)
time.sleep(E_DELAY)

def lcd_string(message,line):
    # Send string to display

    message = message.ljust(LCD_WIDTH," ")

    lcd_byte(line, LCD_CMD)

    for i in range(LCD_WIDTH):
        lcd_byte(ord(message[i]),LCD_CHR)


lcd_init()
lcd_byte(0x01,LCD_CMD)
lcd_string(" WELCOME",LCD_LINE_1)
##str_message=""
##serialPort = serial_rx_tx.SerialPort()
##
##
##def OpenCommand():
##
##    comport = 'COM4'
##    baudrate = '9600'
##    serialPort.Open(comport,baudrate)
##    #serialPort.Close()
##
##
def SendDataCommand(cmd):
    message = str(cmd)
    ser.write(message.encode())
```

```
import telepot
```

```
bot_token = '8570871199:AAFk0O5KLxTQIEoiA46xqUbppVqA-tgUEMk'
ser = serial.Serial ("/dev/ttyS0")
def send_csv_file(file_path, chat_id):
    bot = telepot.Bot(bot_token)
    bot.sendDocument(chat_id, open(file_path, 'rb'))

file_path = 'path/to/your/csv_file.csv'
chat_id = '7808941127' # repla202ce with the chat ID of the recipient

window = tk.Tk()
window.title("Face_Recogniser")

window.geometry('1280x720')
window.configure(background='white')
#window.attributes('-fullscreen', True)
window.grid_rowconfigure(0, weight=1)
window.grid_columnconfigure(0, weight=1)

slist=[1,2,3,4,5]
phlist=['9618393319','9392034450','9959213666','8328229605','9490506568']

message = tk.Label(window, text="FACE ATTENDANCE SYSTEM"
,bg="blue" ,fg="white" ,width=50 ,height=1,font=('times', 20, 'italic bold underline'))

message.place(x=50, y=20)

lbl = tk.Label(window, text="Enter ID",width=16 ,height=2 ,fg="black" ,bg="white"
,font=('times', 15, ' bold '))
lbl.place(x=100, y=100)

txt = tk.Entry(window,width=16 ,bg="white" ,fg="black",font=('times', 15, ' bold '))
txt.place(x=400, y=115)
```

```

lbl2 = tk.Label(window, text="Enter
Name",width=16 ,fg="black" ,bg="white" ,height=2 ,font=('times', 15, ' bold '))
lbl2.place(x=100, y=200)

txt2 = tk.Entry(window,width=16 ,bg="white" ,fg="black",font=('times', 15, ' bold
') )
txt2.place(x=400, y=215)

lbl3 = tk.Label(window, text="Notification :
",width=20 ,fg="orange" ,bg="white" ,height=2 ,font=('times', 15, ' bold underline '))
lbl3.place(x=100, y=300)

message = tk.Label(window, text="....."
,bg="white" ,fg="black" ,width=30 ,height=2, activebackground = "yellow"
,font=('times', 15, ' bold '))
message.place(x=400, y=300)

lbl3 = tk.Label(window, text="Attendance :
",width=20 ,fg="black" ,bg="white" ,height=2 ,font=('times', 15, ' bold underline'))
lbl3.place(x=100, y=550)

message2 = tk.Label(window, text="....."
,fg="black" ,bg="white",activeforeground =
"green",width=30 ,height=2 ,font=('times', 15, ' bold '))
message2.place(x=400, y=550)

bot = telepot.Bot('6812948135:AAG6JbF2MnZnbkReoM8TmV2kaGebs-zmTKI')
def handle(msg):

    global telegramText
    global chat_id
    global receiveTelegramMessage
    global inp
    content_type, chat_type, chat_id = telepot.glance(msg)

```

```
print(content_type)

if content_type == 'text':

    chat_id = msg['chat']['id']
    telegramText = msg['text']
    print("Message received from " + str(chat_id))
    print("Message " + str(telegramText))

bot.message_loop(handle)

def send_sms(x):
    time.sleep(1)
    cmd='AT\r\n'
    SendDataCommand(cmd)
    time.sleep(2)

    cmd='AT+CMGF=1\r\n'
    SendDataCommand(cmd)
    time.sleep(2)

    phno=phlist[x]
    cmd='AT+CMGS="'+str(phno)+"'\r\n'
    SendDataCommand(cmd)
    time.sleep(1)
    cmd="Dear Parent,Your child is absent today"
    SendDataCommand(cmd) # Message
    #ser.write(msg.encode()) # Message
    time.sleep(1)
    cmd = "\x1A"
    SendDataCommand(cmd) # Enable to send SMS
    time.sleep(5)
    print('SMS Sent')
```

```
def clear():
    txt.delete(0, 'end')
    res = ""
    message.configure(text= res)

def clear2():
    txt2.delete(0, 'end')
    res = ""
    message.configure(text= res)

def is_number(s):
    try:
        float(s)
        return True
    except ValueError:
        pass

    try:
        import unicodedata
        unicodedata.numeric(s)
        return True
    except (TypeError, ValueError):
        pass

    return False

def TakeImages():
    Id=(txt.get())
    name=(txt2.get())
    if(is_number(Id) and name.isalpha()):
        cam = cv2.VideoCapture(0)
        harcascadePath = "harcascade_frontalface_default.xml"
        detector=cv2.CascadeClassifier(harcascadePath)
        sampleNum=0
```



```

while(True):
    ret, img = cam.read()
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    faces = detector.detectMultiScale(gray, 1.3, 5)
    for (x,y,w,h) in faces:
        cv2.rectangle(img,(x,y),(x+w,y+h),(255,0,0),2)
        #incrementing sample number
        sampleNum=sampleNum+1
        #saving the captured face in the dataset folder TrainingImage
        cv2.imwrite("TrainingImage/"+name+"."+Id+"."+str(sampleNum)+".jpg",
gray[y:y+h,x:x+w])
        #display the frame
        cv2.imshow('frame',img)
        #wait for 100 milliseconds
        if cv2.waitKey(100) & 0xFF == ord('q'):
            break
        # break if the sample number is morethan 100
        elif sampleNum>60:
            break
    cam.release()
    cv2.destroyAllWindows()
    res = "Images Saved for ID : " + Id + " Name : " + name
    row = [Id , name]
    with open('StudentDetails/StudentDetails.csv','a+') as csvFile:
        writer = csv.writer(csvFile)
        writer.writerow(row)
    csvFile.close()
    message.configure(text= res)
else:
    if(is_number(Id)):
        res = "Enter Alphabetical Name"
        message.configure(text= res)
    if(name.isalpha()):
        res = "Enter Numeric Id"

```

```
message.configure(text= res)

def TrainImages():
    recognizer = cv2.face_LBPHFaceRecognizer.create()#recognizer =
cv2.face.LBPHFaceRecognizer_create()#$cv2.createLBPHFaceRecognizer()
    harcascadePath = "haarcascade_frontalface_default.xml"
    detector =cv2.CascadeClassifier(harcascadePath)
    faces,Id = getImagesAndLabels("TrainingImage")
    recognizer.train(faces, np.array(Id))
    recognizer.save("Trainer.yml")
    res = "Image Trained"#+",".join(str(f) for f in Id)
    message.configure(text= res)

def getImagesAndLabels(path):
    #get the path of all the files in the folder
    imagePaths=[os.path.join(path,f) for f in os.listdir(path)]
    #print(imagePaths)

    #create empty face list
    faces=[]
    #create empty ID list
    Ids=[]
    #now looping through all the image paths and loading the Ids and the images
    for imagePath in imagePaths:
        #loading the image and converting it to gray scale
        pilImage=Image.open(imagePath).convert('L')
        #Now we are converting the PIL image into numpy array
        imageNp=np.array(pilImage,'uint8')
        #getting the Id from the image
        Id=int(os.path.split(imagePath)[-1].split(".")[1])
        # extract the face from the training image sample
        faces.append(imageNp)
        Ids.append(Id)
    return faces,Ids
```

```

def TrackImages():
    global slist

    recognizer =
cv2.face.LBPHFaceRecognizer_create()#cv2.createLBPHFaceRecognizer()
    recognizer.read("Trainer.yml")
    harcascadePath = "haarcascade_frontalface_default.xml"
    faceCascade = cv2.CascadeClassifier(harcascadePath);
    df=pd.read_csv("StudentDetails/StudentDetails.csv")
    cam = cv2.VideoCapture(0)
    font = cv2.FONT_HERSHEY_SIMPLEX
    col_names = ['Id','Name','Date','Time']
    attendance = pd.DataFrame(columns = col_names)
    outtime=0
    intime=0
    while True:
        ret, im =cam.read()
        gray=cv2.cvtColor(im,cv2.COLOR_BGR2GRAY)
        faces=faceCascade.detectMultiScale(gray, 1.2,5)
        for(x,y,w,h) in faces:
            cv2.rectangle(im,(x,y),(x+w,y+h),(225,0,0),2)
            Id, conf = recognizer.predict(gray[y:y+h,x:x+w])
            if(conf < 70):
                ts = time.time()
                date = datetime.datetime.fromtimestamp(ts).strftime('%Y-%m-%d')
                timeStamp = datetime.datetime.fromtimestamp(ts).strftime('%H:%M:%S')
                aa=df.loc[df['Id'] == Id]['Name'].values
                tt=str(Id)+"-"+aa
                hour = int(strftime("%H"))
                mint = int(strftime("%M"))
                sec = int(strftime("%S"))
                print (str(hour) + ":" + str(mint) + ":" + str(sec))
                sleep(1)

```

```

if(hour == a1_h and mint == a1_m and sec<10):
    intime=1
    print('starting')
    GPIO.output(buz,0)
    time.sleep(1)
    GPIO.output(buz,1)
if(hour == a2_h and mint == a2_m and sec<10):
    intime=0
    print('ending')
    GPIO.output(buz,0)
    time.sleep(1)
    GPIO.output(buz,1)
if(hour == a3_h and mint == a3_m and sec<10):
    outtime=1

if(intime==1):
    attendance.loc[len(attendance)] = [Id,aa,date,timeStamp]
    if(Id in slist):
        slist.remove(Id)
    print(slist)
    # lcd_byte(0x01,LCD_CMD)
    # lcd_string("S:" + str(attendance),LCD_LINE_1)
#lcd_string("T:" + str(tval),LCD_LINE_2)

else:
    Id='Unknown'
    tt=str(Id)
    if(conf > 75):
        noOfFile=len(os.listdir("ImagesUnknown"))+1
        cv2.imwrite("ImagesUnknown/Image"+str(noOfFile) + ".jpg",
im[y:y+h,x:x+w])
        cv2.putText(im,str(tt),(x,y+h), font, 1,(255,255,255),2)
        lcd_byte(0x01,LCD_CMD)

```

```

        lcd_string(str(tt),LCD_LINE_1)
        attendance=attendance.drop_duplicates(subset=['Id'],keep='first')
        cv2.imshow('im',im)
        if (cv2.waitKey(1)==ord('q') or outtime==1):
            break
    print('sending info')
    ts = time.time()
    date = datetime.datetime.fromtimestamp(ts).strftime('%Y-%m-%d')
    timeStamp = datetime.datetime.fromtimestamp(ts).strftime('%H:%M:%S')
    Hour,Minute,Second=timeStamp.split(":")
    fileName="Attendance/Attendance_"+date+"_"+Hour+"-"+Minute+"-
"+Second+".csv"
    attendance.to_csv(fileName,index=False)
    cam.release()
    cv2.destroyAllWindows()
    #print(attendance)
    res=attendance
    message2.configure(text= res)
    send_csv_file(fileName, chat_id)
    for x in range(len(slist)):

        print ('Sending SMS To'+ str(slist[x]))
        send_sms(slist[x])
        time.sleep(0.1)

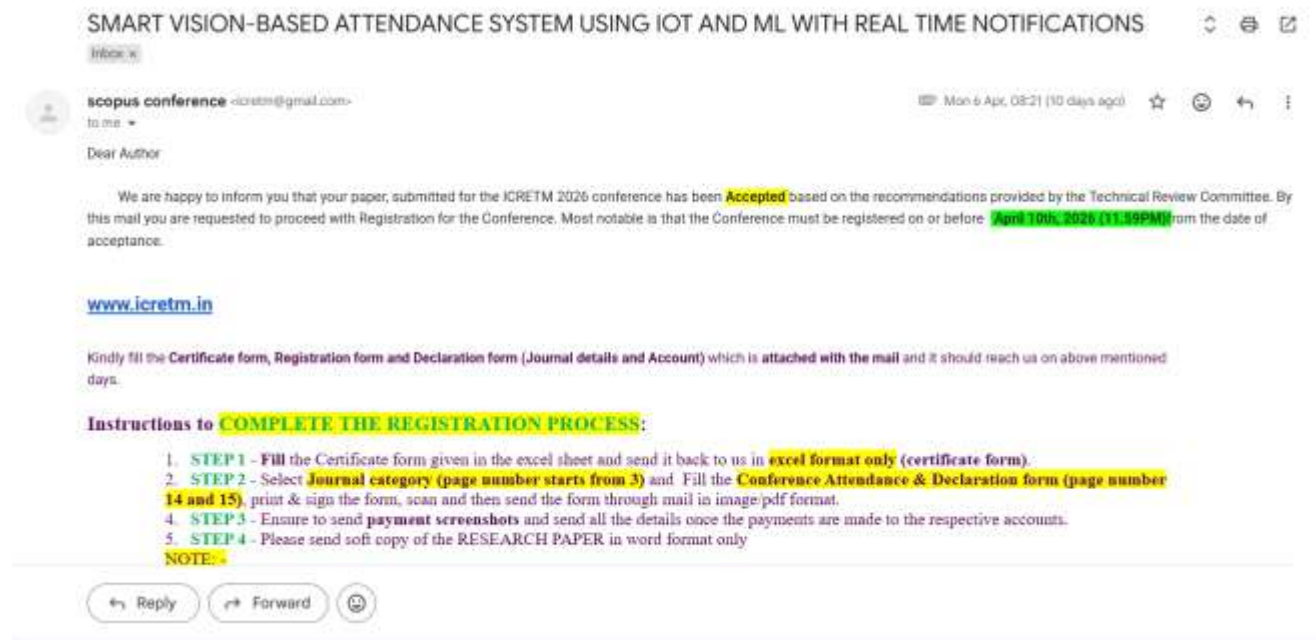
clearButton = tk.Button(window, text="Clear",
command=clear ,fg="black" ,bg="gray" ,width=10 ,height=1 ,activebackground =
"Red" ,font=('times', 15, ' bold '))
clearButton.place(x=650, y=100)
clearButton2 = tk.Button(window, text="Clear",
command=clear2 ,fg="black" ,bg="gray" ,width=10 ,height=1, activebackground =
"Red" ,font=('times', 15, ' bold '))
clearButton2.place(x=650, y=200)

```

```
takeImg = tk.Button(window, text="Take Images",
command=TakeImages ,fg="black" ,bg="gray" ,width=10 ,height=2,
activebackground = "Red" ,font=('times', 15, ' bold '))
takeImg.place(x=100, y=400)
trainImg = tk.Button(window, text="Train Images",
command=TrainImages ,fg="black" ,bg="gray" ,width=10 ,height=2,
activebackground = "Red" ,font=('times', 15, ' bold '))
trainImg.place(x=300, y=400)
trackImg = tk.Button(window, text="Track Images",
command=TrackImages ,fg="black" ,bg="gray" ,width=10 ,height=2,
activebackground = "Red" ,font=('times', 15, ' bold '))
trackImg.place(x=500, y=400)
quitWindow = tk.Button(window, text="Quit",
command=window.destroy ,fg="black" ,bg="gray" ,width=10 ,height=2,
activebackground = "Red" ,font=('times', 15, ' bold '))
quitWindow.place(x=700, y=400)

window.mainloop()
```

APPENDIX B



Conference Paper Acceptance Email – ICRETM 2026

CERTIFICATES







SMART VISION-BASED ATTENDANCE SYSTEM USING IOT AND ML WITH REAL TIME NOTIFICATIONS

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Abstract— Attendance management is an essential administrative task in educational institutions; however, traditional attendance methods such as manual registers and RFID cards require more time, allow proxy attendance and do not provide proper real-time monitoring. This paper presents a Smart Vision-Based Attendance System integrating Internet of Things (IoT) and Machine Learning (ML) to automate the process. In this system, a camera captures live video and a face recognition algorithm is used to detect and identify registered students automatically. Attendance records are saved in a central database and real-time notifications are sent to parents in case of student absence through a communication module. The proposed system reduces human intervention and provides a scalable and efficient solution for smart classrooms.

Index Terms— Vision-Based Attendance System, Face Recognition, Machine Learning (ML), Local Binary Pattern Histogram (LBPH), Internet of Things (IoT), Real-Time Notification System.

I. INTRODUCTION

Attendance management is an important task in educational institutions and organizations because it helps monitor student participation and academic discipline, as it plays a vital role in academic evaluation, discipline monitoring and administrative planning. Traditional methods, like manual roll calls or register-based marking, are still commonly used. However, these approaches are time-consuming, inefficient for large classrooms and highly dependent on human involvement, which often leads to errors and data inconsistency [1].

Recent progress in computer vision and machine learning have enabled the development of smart systems that can recognize individuals based on their facial features. Face recognition technology is widely used because it can identify people easily without physical contact and provides higher accuracy than traditional biometric methods [3]. Vision-based systems can detect and recognize faces from live video streams, making them suitable for smart classrooms and surveillance

applications. Integrating Internet of Things (IoT) with vision-based systems allows attendance data to be monitored and stored in the cloud. This makes it possible to access attendance records anytime and from any location. It also enables automatic notifications to inform parents about student attendance. The system also reduces mistakes and gives real-time updates to parents.

However, many existing vision-based attendance systems still have limitations such as limited scalability, lack of automatic alerts and dependence on local data storage. In many institutions, delayed information about student absence reduces transparency and prevents quick action. Therefore, a smart and contactless attendance system is needed.

This paper proposes a Smart Vision-Based Attendance System using IoT and Machine Learning. The system uses a camera to capture images and identify students using face recognition. Attendance is recorded automatically and stored in a cloud database. Real-time notifications are sent when a student is absent. The system reduces manual work, prevents proxy attendance and provides an efficient solution for modern classrooms.

II. EXISTING METHODS

Attendance management systems have evolved from manual approaches to automated and intelligent solutions. Existing methods can be broadly classified into **manual systems, card-based systems, biometric systems, and vision-based systems.**

A. Manual Attendance Systems

Manual attendance systems use roll calls or registers to record attendance. The teacher marks each student manually. These methods are easy to use but take more time and may cause human errors. They also do not support digital storage or real-time monitoring[1].

B. Card-Based Attendance Systems

Card-based systems use RFID cards or smart ID cards to record attendance. Each student scans a card to mark their presence. Although this method reduces manual work, proxy attendance is

still possible if one student uses another student's card[2].

C. Biometric Attendance Systems

Biometric systems identify individuals using physical features such as fingerprints or iris patterns. These systems provide better accuracy than card-based systems. However, they require physical contact and regular maintenance of sensors[3].

D. Vision-Based Attendance Systems

Vision-based attendance systems use cameras and face recognition algorithms to detect and identify individuals. These systems allow contactless attendance marking and reduce proxy attendance. However, some existing systems lack cloud storage and real-time notification features[4].

E. Limitations of Existing Methods

Existing attendance systems need a lot of manual work, can be cheated with proxy attendance and do not give real-time updates. They are also hard to use for large groups. A smart system using face recognition, IoT and cloud storage can fix these problems. It saves time, lowers mistakes, and gives instant attendance updates.

III. PROPOSED SYSTEM

A. System Overview

The proposed system is a **Smart Vision-Based Attendance Management framework** designed to automate attendance recording using real-time facial recognition. The system integrates **camera-based image acquisition, machine learning-based face recognition, and centralized cloud storage** to ensure accurate and secure attendance monitoring, following intelligent automation principles [1], [2].

Attendance is permitted only for **pre-registered users** during predefined time intervals. Individuals are marked present only after **successful facial verification** against the stored database. The recorded attendance data is updated in real time on the central server and **instant notifications** are generated for authorized users, enabling efficient and reliable attendance management.

B. Block Diagram

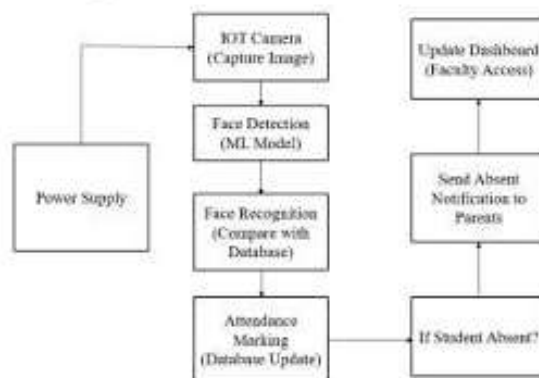


Fig.1. Block Diagram Of Smart Vision-Based Attendance System

The block diagram shows a layered architecture of the proposed vision-based attendance system, where each layer performs a

specific task for accurate and automated attendance management.

Image Acquisition (IoT Layer): A camera captures live images or video frames of students and sends the data to the next stage.

Real-Time Face Recognition (Machine Learning Layer): LBPH-based algorithms detect and recognize faces by comparing extracted features with the registered dataset. Only known individuals are identified to ensure accuracy.

Attendance Verification and Storage (Database Layer): Recognized faces are verified to prevent duplicate entries and securely stored in the database with date and time information.

Notification and Reporting (Communication Layer): Attendance reports are generated and notifications about absences are sent via email or messaging services.

Faculty Dashboard Access (Application Layer): Authorized users can view attendance records and reports through a user-friendly interface, enabling real-time monitoring and management.

This layered design ensures smooth data flow from image capture to reporting, providing a fully automated, scalable, and efficient attendance system.

C. Software Architecture

The software for the Smart Vision-Based Attendance System is built to continuously capture images, identify faces instantly, mark attendance automatically and send alerts. It is organized in layers to make the system dependable, easy to expand and simple to maintain.

1. Data Acquisition Layer

This layer captures facial images using the camera during the attendance session. The captured images are sent to the processing module for further analysis.

2. Data Processing Layer

In this layer, the system detects faces and extracts facial features using machine learning algorithms. These features are compared with the stored dataset to identify the person.

3. Decision and Control Layer

This layer decides whether the detected person is a registered user. If a match is found, the system marks attendance and avoids duplicate entries.

4. User Interface and Display Layer

The user interface layer provides **real-time attendance feedback** to administrators through a monitoring dashboard. Attendance status, recognition results and system performance indicators are displayed dynamically on the dashboard. This layer eliminates manual verification and enables efficient supervision of attendance activities.

5. Alert and Notification Layer

This layer generates **real-time notifications** upon successful attendance marking or failed recognition attempts. Alerts are transmitted to authorized users through messaging or notification services, ensuring transparency and immediate awareness without manual intervention.

6. Data Logging and Storage Layer

The system maintains logs of **attendance records, recognition events, timestamps, and notification triggers** in a centralized database. These records support report generation, performance analysis and future system optimization. Historical attendance data also enable trend analysis and system reliability evaluation. The system stores all attendance details, recognition events, time stamps, and notification actions in a central database. These records allow easy generation of reports and analysis of system performance. Past attendance data can be used to identify patterns and trends over time.

IV. METHODOLOGY

The proposed methodology describes the process to design and implement a vision-based attendance system using IoT, machine learning, and cloud services. It ensures accurate, automated, and reliable attendance recording.

A. System Initialization

All modules, including the camera, machine learning model, cloud database and notification services, are initialized. Pre-trained facial datasets of registered individuals are loaded and network connectivity is established for seamless cloud communication.

B. Image Acquisition

The camera continuously captures live video frames or images from the classroom. Frames are sent to the processing unit for further analysis.

C. Face Detection

Each captured frame undergoes face detection to locate human faces while ignoring background objects. Only detected facial regions are processed further, which improves efficiency and reduces unnecessary computation.

The algorithm based on Local Binary Patterns and histograms (LBPH) is used for face recognition because it is simple, reliable and requires low computational resources. LBPH extracts local texture features from facial images by comparing each pixel with its neighbors to generate a binary pattern. The image is divided into smaller regions, and histograms of these patterns are created for each region. All regional histograms are combined to form a single feature set representing the face. During recognition, this feature set is compared with stored datasets using similarity measures, and the closest match determines the person's identity.

LBPH works well under different lighting conditions, requires minimal training data, and is suitable for real-time face recognition on devices like Raspberry Pi or personal computers. This makes it an efficient and affordable solution for attendance tracking and smart surveillance systems.

D. Face Recognition Using Machine Learning

Unique facial features are compared with stored datasets. A match confirms the identity, while unmatched faces are marked as unknown.

E. Attendance Marking

Confirmed identities are automatically recorded with date and time. Duplicate entries are avoided for the same session.

F. Cloud Data Storage

Attendance records are sent to a cloud database for secure, centralized storage and remote access. This removes manual record-keeping and supports scalability.

G. Absence Detection and Notification

After each session, unrecognized individuals are marked absent. Automated notifications are sent to parents, guardians, or administrators for timely awareness.

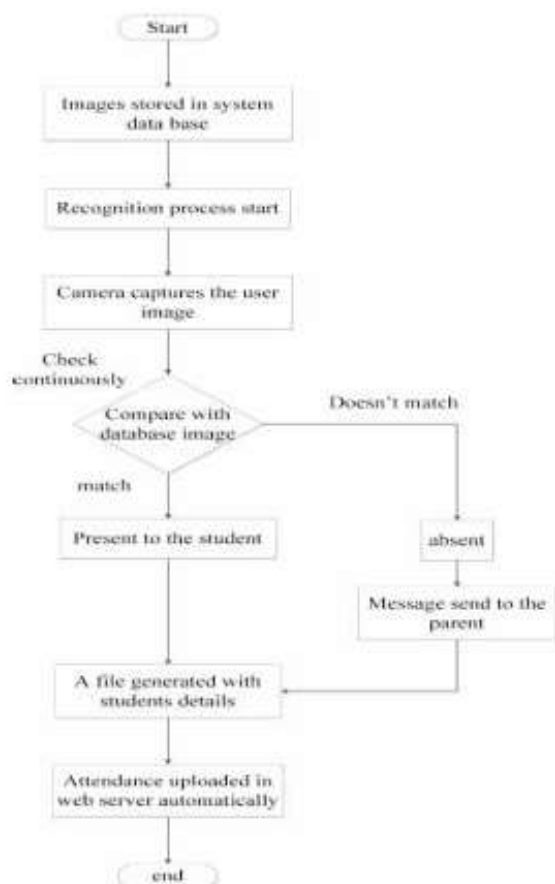
H. Data Access and Monitoring

Authorized users can view attendance records and reports via a mobile or web interface, helping monitor patterns and trends.

I. Methodology Summary

The system combines IoT, cloud computing, computer vision, and machine learning to provide a fully automated, contactless, and scalable attendance solution. Each stage ensures accurate attendance, secure data handling, and real-time notifications.

V. FLOWCHART



VI. HARDWARE IMPLEMENTATION

The system's hardware consists of a camera, processing unit, storage, network interface, and power supply. Together, these components allow the system to capture faces, recognize individuals and automatically record attendance in real time with cloud support.

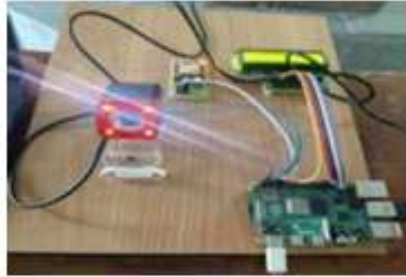


Fig.2.Hardware Implementation of the Proposed System

A. Camera Module

The camera captures live images of students, ensuring clear visibility of faces. Captured frames are sent to the processor for face detection and recognition. A USB or Raspberry Pi camera can be used.



Fig.3. Web Camera

B. Processing Unit (Raspberry Pi)

The processing unit runs face detection, recognition, and attendance management. It extracts facial features and compares them with the stored dataset. Raspberry Pi is chosen for its compact size, low power, and compatibility with Python ML libraries.



Fig.4.Raspberry Pi 4 Model B Board

C. Memory and Storage

Captured images, facial datasets, and processing results are temporarily stored locally before attendance records are sent to the cloud for permanent storage.

D. Cloud Platform Interface

The cloud provides centralized storage and real-time access. It supports data analysis and allows the system to scale for multiple classrooms or institutions.

E. Notification Interface

Alerts for absenteeism are sent via cloud-based messaging or email. The hardware supports communication through modules like GSM to notify parents or administrators.

F. Power Supply Unit

The power unit ensures stable electricity to all components, preventing interruptions and protecting hardware from voltage fluctuations.

VII. USER INTERFACE DESIGN

The front-end of the proposed attendance system has a graphical user interface (GUI) for registration, training, and attendance management. It is designed to be simple and reliable.

Users enter a unique ID and name to label facial images during dataset creation. The "Take Images" option captures multiple samples for training. The "Train Images" button extracts facial features and creates a recognition model for real-time tracking.

The "Track Images" option activates live recognition. The system detects faces, recognizes registered users, and automatically marks attendance, displaying the results on the interface and uploading data to the cloud.

The GUI also shows notifications for status updates, such as image capture, training completion, and attendance confirmation. "Clear" resets fields, and "Quit" closes the application safely.

Overall, the interface enables easy face registration, model training, and attendance tracking, ensuring smooth interaction with the backend system..

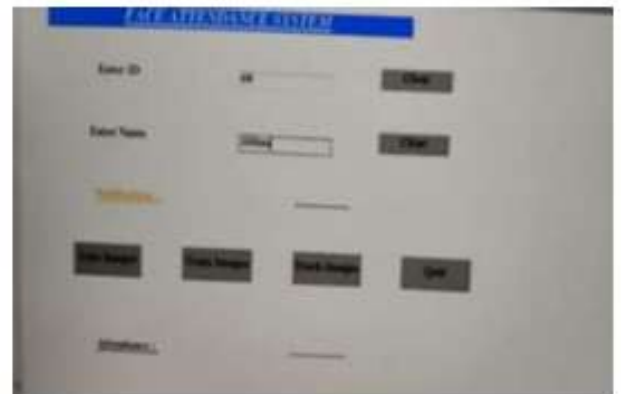


Fig.6.User Interface of the Attendance System

VIII. RESULTS AND DISCUSSION

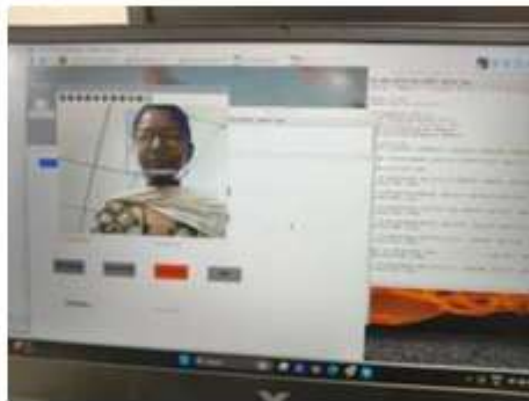
The developed vision-based attendance system was experimentally tested in a classroom-like setting. Four registered individuals — Sowmya, Abbas, Srikanth and Bhaviya — were

enrolled in the facial dataset. Attendance was recorded automatically using face recognition, cloud storage, and notification mechanisms.

A. Student Facial Image Dataset



B. Real Time Face Recognition Output Identifying a Registered User



(a) Registered User 1



(b) Registered User 2

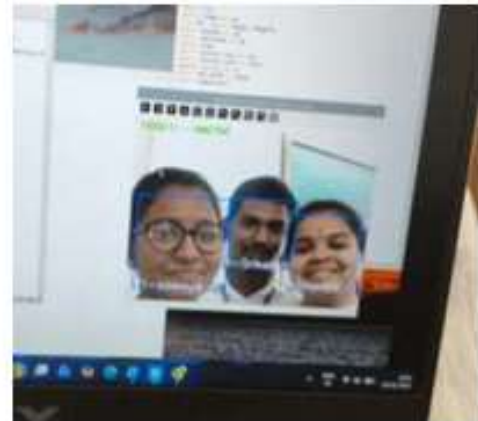


(c) Registered User 3

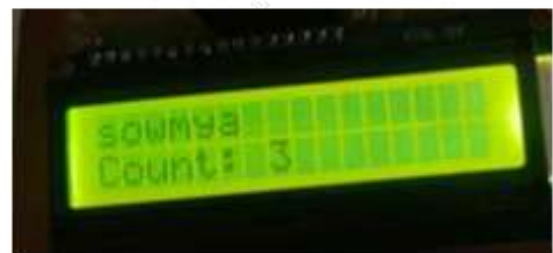


(d) Registered User 4

G. Real-Time Recognition of Multiple Registered Users in a Single Frame



H. Real Time User Recognition and Head Count



I. Automated Attendance Log Stored in Excel Format

	A	B	C	D	E
1	Id	Name	Status	Time	Date
2	1	sowmya	Present	17:47:16	12-02-2026
3	2	Bhaviya	Present	17:47:17	12-02-2026
4	3	Abbas	Absent	N/A	12-02-2026
5	4	Srikanth	Present	17:47:18	12-02-2026
6					
7					
8					

J. Real-Time SMS Notification Sent to Parents



The IoT and machine learning-based attendance system was tested in a real classroom. It captured facial images, trained the recognition model, detected faces in real time, and marked attendance automatically.

During registration, the system stored multiple facial samples for each user, and notifications confirmed successful image capture.

The model trained correctly and worked reliably during repeated tests.

In real-time use, registered faces were recognized accurately, attendance was recorded with date and time, and duplicates were avoided. Unknown faces were ignored, and absentees were marked correctly.

The results show the system is accurate, reduces manual work, prevents proxy attendance, and allows secure cloud access with notifications to keep administrators informed.

IX. FUTURE SCOPE

The proposed smart vision-based attendance system can be further improved using advanced technologies to enhance accuracy, intelligence, and scalability.

A. Deep Learning Based Multi-Angle Recognition

Apply neural networks to recognize faces from multiple angles, even in crowded classrooms.

B. Student Attention and Behavioral Analysis

Use facial expression detection to monitor attention and engagement.

C. Edge Computing for Real-Time Processing

Process video locally for faster, secure, and low-latency attendance marking.

D. Integration with Smart Campus Ecosystems

Connect with other IoT-enabled campus systems for a fully automated smart ecosystem.

X. CONCLUSION

This paper presents a Smart Vision-Based Attendance System using IoT and machine learning for automatic classroom attendance. It is contactless, fast, and accurate, reducing manual work and errors. Cloud storage keeps data secure and easily accessible, while real-time notifications **inform guardians and administrators** about absences. Tests show the system performs reliably under different classroom conditions. Future

improvements may include deep learning, mobile apps and multi-camera support for larger classrooms.

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REFERENCES

- [1] N. Thaleparambil, A. Biju, and B. Prathap, "Integrated automated attendance system with RFID, Wi-Fi, and visual recognition technology for enhanced classroom security and precise monitoring," in *Proc. IEEE Int. Conf. on Contemporary Computing and Communication (InC4)*, 2024.
- [2] D. V. Darshan *et al.*, "IoT-based attendance monitoring with facial recognition," in *Proc. IEEE Int. Conf. on Advanced Information Technology (ICAIT)*, 2024.
- [3] A. Raj *et al.*, "Facial recognition-based student attendance system," in *Proc. IEEE Int. Conf. on Recent Innovations in Technology and Optimization (ICRITO)*, 2024.
- [4] G. Ramani *et al.*, "Smart attendance system using IoT," in *Proc. IEEE Int. Conf. on Communication and Computational Systems (ICCCS)*, 2023.
- [5] N. Ramakrishnan, A. Ajil, A. Ali, A. M. Sundaram, and V. Sanju, "Wi-Fi-based smart attendance monitoring system," in *Proc. IEEE Conf. on Communication Systems, Computing and IT Applications (CSITSS)*, 2023.
- [6] O. Kale, D. Kharat, S. Babar, and C. Barse, "Real-time attendance system using facial recognition and web-based integration," *Int. J. Creative Res. Thoughts*, vol. 13, no. 3, pp. 1–8, Mar. 2025.
- [7] D. K. B., D. C. N., R. H. V., S. N., and T. P. M., "Real-time student face recognition attendance system using AI," *Int. Adv. Res. J. Sci. Eng. Technol.*, vol. 12, no. 5, pp. 1067–1072, May 2025.
- [8] M. Naveen Raj and R. Vadivel, "Face recognition-based attendance system using machine learning with location identification," *World J. Adv. Res. Rev.*, vol. 18, no. 1, pp. 1029–1035, 2023.
- [9] T. Abderraouf, A. A. Wassim, and S. Larabi, "An embedded intelligent system for attendance monitoring," *arXiv preprint*, Jun. 2024.
- [10] S. Das and A. Mukherjee, "Smart classroom attendance management using AI and IoT with real-time alerts," *Int. J. Eng. Res. Technol.*, vol. 15, no. 9, pp. 1182–1190, Sep. 2024.